Physics of the Earth and Planetary Interiors, 24 (1981) 142–168
Elsevier Scientific Publishing Company, Amsterdam – Printed in The Netherlands

ASPECTS OF PRECAMBRIAN PALAEOMAGNETISM, WITH NEW DATA FROM THE LIMPOPO MOBILE BELT AND KAAPVAAL CRATON IN SOUTHERN AFRICA

G.E. MORGAN * and J.C. BRIDEN

Department of Earth Sciences, University of Leeds, Leeds LS2 9JT (Great Britain)

(Received March 20, 1979; accepted for publication May 5, 1980)

Morgan, G.E. and Briden, J.C., 1981. Aspects of Precambrian palaeomagnetism, with new data from the Limpopo mobile belt and Kaapvaal craton in southern Africa. Phys. Earth Planet. Inter., 24: 142–168.

Palaeomagnetic results from Precambrian rocks suggest that the geomagnetic field in the Precambrian was similar in major respects to the Recent field, although there is a possibility that reversals were less frequent. Limitations of precision in Proterozoic palaeomagnetic and geochronological data, and inadequate geographic spread of these data, prevent the making of complete Proterozoic continental reconstructions. Nevertheless, where the data are most abundant, Precambrian apparent polar wander (APW) paths can be used in determining the relative movement of cratons and groups of cratons. Relative movement between Africa, Laurentia and Australia during at least part of the Proterozoic seems to be indicated, but the lack of significant differences between the APW paths of several pairs of presently adjacent Proterozoic cratons suggests that *some* Proterozoic orogenic belts were not formed by the collision of previously widely separated blocks. Palaeomagnetic studies on metamorphic and igneous rocks from within Proterozoic orogenic belts, notably the Grenville belt in Canada, show that the effects of slow post-orogenic cooling are commonly detectable palaeomagnetically.

The problems and potential of Precambrian palaeomagnetism are illustrated by studies on rocks from the Kaapvaal craton and adjacent Limpopo mobile belt in southern Africa, and new data are presented from (i) a stratigraphically important horizon in the Waterberg Supergroup, (ii) the Palabora igneous complex and (iii) the metamorphic rocks of the Limpopo belt. The results from the Palabora complex indicate two distinct bursts of igneous activity at ca. 2000 and ca. 1900 My. The rocks of the Limpopo belt have two stable magnetizations, these sometimes residing together in the same specimens and resolvable only by vector analysis of detailed demagnetization data. Component A is thought to have been acquired during cooling of the belt at about 2000 My, and this result, together with one previously published, suggests diachronous cooling across the belt with a polarity change occurring during the cooling interval. Component B is thought to be a later CRM but its exact age is unknown. The new data assist in refining the African APW path for the interval ca. 2000 to ca. 1850 My. This segment of the African APW path is shown to have a distinctive polarity pattern, and a similar pattern is also recognised in the Laurentian APW path for this interval. These polarity patterns help to resolve the ambiguity of geographical polarity to be assigned to these APW paths and, following from this, evidence is presented which suggests that Africa and Laurentia were in relative motion at about 1950 My.

1. Introduction

In the first decades of large scale acquisition of palaeomagnetic data in the 1950s and 1960s, efforts were concentrated worldwide on the Mesozoic and Tertiary at the expense – with a few notable exceptions – of the Palaeozoic and especially the Precambrian. The prime reason for this course of development was that the corpus of knowledge about the past geomagnetic field and on past continental distributions was necessarily best built up step by step, working backwards from the geologically recent. Moreover, it could be anticipated that a variety of limiting factors and interpretational problems which existed in palaeomagnetic study of relatively young rocks, would be compounded as one ventured further

* Present address: Exploration Department, Gulf Canada Resources Inc., P.O. Box 130, Calgary, Alberta, T2P 2H7, Canada.

0031-9201/81/0000–0000/$02.50 ©1981 Elsevier Scientific Publishing Company

back in time. Thus the definition of age of origin of NRM is potentially inhibited by the uncertainties in stratigraphic age assignment, by the experimental errors associated with radiometric age determination, and by uncertainties as to whether the age of the NRM need necessarily coincide with the stratigraphic or radiometric age; and all these problems increase with the age of the rock units being investigated. By virtue of their age per se, Precambrian rocks are more likely than younger rocks to have been subjected to secondary effects. In particular, the original NRM may have been completely or partially reset by later metamorphic events, and, like some mineral isotope systems, may even have been reset by a thermal event so mild that it produced no appreciable petrographic effects. In many Precambrian terrains metamorphism and deformation have been widespread and often intense, so that remagnetization, superposition of NRM components of various ages, or distortion of NRM due to anisotropy of stress or rock fabric are all to be anticipated.

Thus it is not surprising that before progressive demagnetization techniques became routine in the early 1960s the few palaeomagnetic surveys of Precambrian rocks were confined almost exclusively to scarcely deformed and scarcely metamorphosed supracrustal sediments and volcanics (nearly all 1200 My old or less) which had been laid down on already established Archaean–Proterozoic cratons in Europe, North America, South America, China and Australia, together with a few studies of unmetamorphosed igneous rocks intruding those terrains. On this basis, Precambrian studies were simply an extension back in time of the approach adopted by most workers in Phanerozoic palaeomagnetism, whose motivations were on a global scale in both geomagnetism (i.e. configuration of the main field and its reversal property) and geotectonics (in the form of reconstruction of past continental configurations).

The only element of novelty in these early studies of Precambrian rocks was the investigation of some gneisses (Balsley and Buddington, 1957) which indicated that consistent magnetizations did exist in some metamorphic rocks. The cue which was thus given to further study of metamorphic terrains was little taken up until about 1970 – perhaps because of the variety of difficulties that were in prospect – and even now remains well short of fulfilling its main potential. This theme will be further developed in Sections 4 and 5, and the view is expressed that the future potential for palaeomagnetic investigation of thermal history of metamorphic belts is very great.

2. Precambrian geomagnetic field

Perhaps the most obvious points to emerge from even the early studies of Precambrian palaeomagnetism were those with geomagnetic implications. Studies of individual rock units produced results which were identical in major features with those from Phanerozoic to Recent rocks. Reversals of magnetization were traced stratigraphically in ca. 760 My old sediments of the Torridon Group in northwest Scotland (Irving and Runcorn, 1957). As with younger polarity zones, 180° reversals appeared to have been achieved. A polarity transition sequence in ca. 1700 My old red sediments in the Great Slave area of Canada (Bingham and Evans 1975, 1976) was found to be similar in all respects to more recent transitions, although its duration might possibly have been longer. Opposite polarities were found in the Modipe Gabbro in Botswana (Evans and McElhinny, 1966) with which is associated a radiometric age of 2630 ± 470 My. Agreement of mean directions of magnetization, regardless of polarity, between rocks of similar Proterozoic ages in the same structural province, and sequential changes of magnetization directions with time are observed. These lead to apparent polar wander (APW) paths for large regions of Precambrian continents which are similar in style to Phanerozoic APW paths (e.g. Piper, 1976a; McElhinny and McWilliams, 1977; McGlynn and Irving, 1978; see Fig. 1). Dispersion of stable NRM directions within individual formations, which in favourable circumstances are interpreted as measures of palaeo-secular variation, are comparable to those in younger rocks (McElhinny et al., 1968).

All these features are indicative of a Precambrian geomagnetic field identical in major respects to the Recent geomagnetic field; i.e. a predominantly dipole field which repeatedly reversed (sufficiently quickly, or rarely, so as not often to be observed in its transitional state) and which during constant polarity was subject to secular variations of several degrees about its mean. Moreover the continental crust appears to

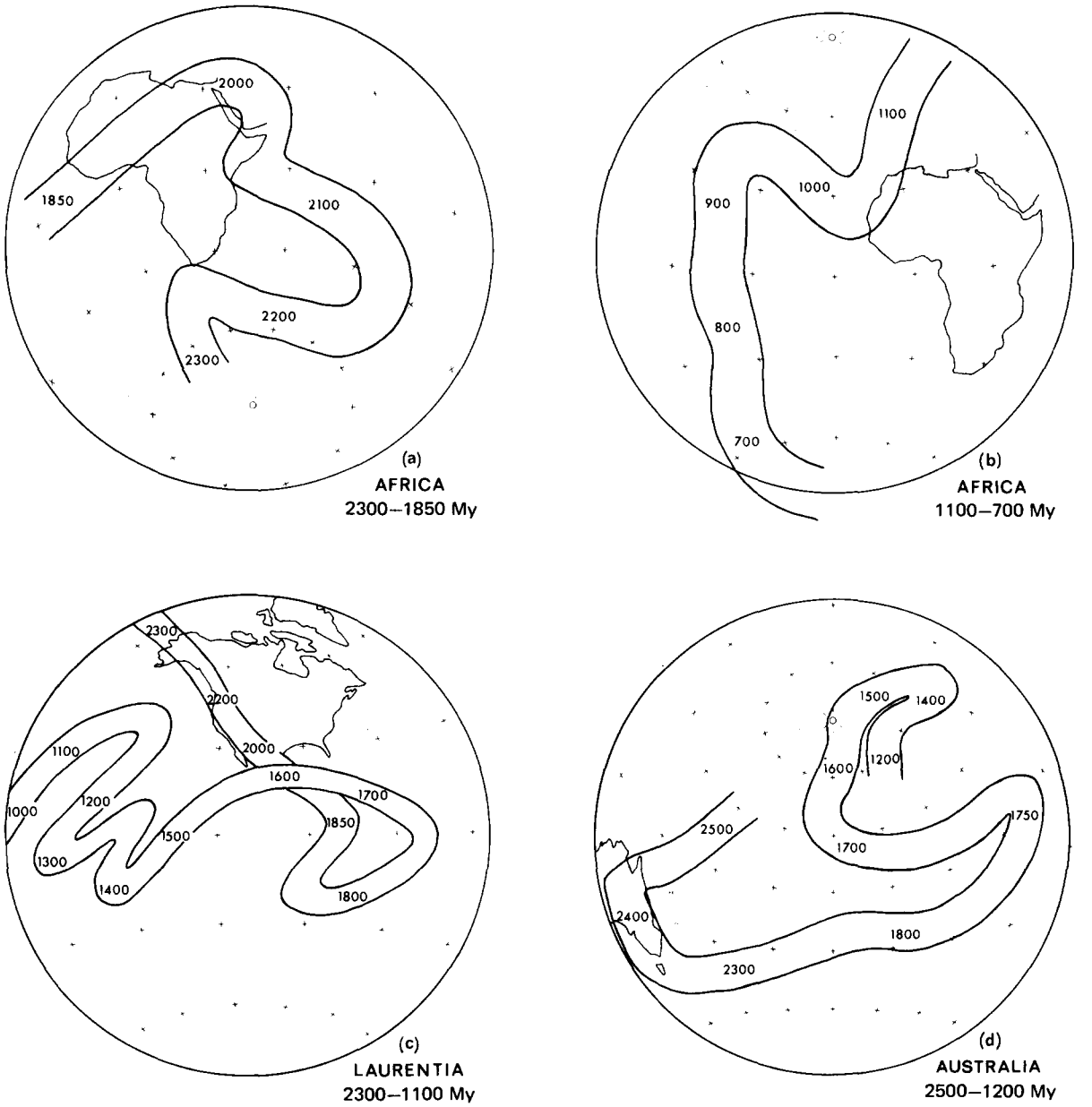


Fig. 1. Some schematic examples of Precambrian APW paths. (a) After McElhinny and McWilliams (1977) with modifications. (b) and (c) after McElhinny and McWilliams (1977). (d) after McGlynn and Irving (1978).

have been drifting relative to this field at much the same rate as in Phanerozoic time.

The only feature that we are aware of that seems to be different from the more recent field is that the occurrence of both polarities in individual rock units in the Precambrian appears to be unusually low. Thus out of approximately 500 Precambrian palaeomagnetic results listed by Irving and Hastie (1975) only

11% are of mixed polarity, whereas for approximately 2400 Phanerozoic results listed by Irving et al. (1976a, b, c) the corresponding figure is 27%. However, when the Phanerozoic data are examined in detail it is seen that there is a progressive decrease in the proportion of mixed polarity results going back in time; Cenozoic 38%, Mesozoic 27% and Palaeozoic 18%, so that the Precambrian was simply the start of a gradual change which continued throughout the Phanerozoic. Irving and Pullaiah (1976) showed that the Phanerozoic was characterized by an alternation of quiet intervals with infrequent reversals and disturbed intervals with frequent reversals. The change in the proportion of mixed polarity results with time described above could be an indication that reversal frequency has gradually increased during Earth history, perhaps due to a gradual increase in the proportion of disturbed intervals. Irving and Pullaiah also showed that most of the Phanerozoic was characterized by long intervals of polarity bias. If this has been a general feature of the geomagnetic field then the gradually decreasing proportion of mixed polarity results going back in time could alternatively be explained as simply a reflection of the fact that in older rock units there is a gradually increasing proportion of those whose remanence was slowly acquired and which records only the overall bias of the field during the magnetization interval. Such time-averaged magnetization might be acquired during very slow cooling of deeply buried metamorphic and igneous terrains, or during very slow chemical growth of remanence carriers. The question of polarity bias and the time-averaging effects of slow magnetization in metamorphic terrains is discussed further in Sections 4 and 5.

3. Palaeomagnetism of cratonic rocks

3.1. Continental reconstructions

The ideal palaeomagnetic approach to palaeocontinental reconstruction is to compile APW paths for each and every tectonic block of interest. In this approach one only uses data for a particular span of time, from rocks which belonged to the particular tectonic block during that time. Thus, for example, one would only use Indian palaeomagnetic data in an analysis of Asia for the last few tens of My during which India has been integral to Asia; likewise one could potentially use East Antarctic palaeomagnetic data to enhance the Australian palaeomagnetic record for that part of pre-Tertiary time for which a fixed contiguity of Australia and East Antarctica is established.

There are still completely inadequate data to compare APW paths for individual Proterozoic blocks in this manner. The unevenness of data, and particularly the different "instants" of the record which have been observed in different places, will ensure that all such APW paths will look different and this could lead to the prima facie, but utterly ill-founded, conclusion that all such blocks have moved independently like the present-day continents (Burke et al., 1976).

That is not to say that there are not other, more satisfactory, ways of comparing global Proterozoic palaeomagnetic data. Confronted with similar problems of data inadequacy in the Phanerozoic, Briden et al. (1973) and McElhinny (1973) used geologically and geometrically based reassemblies for which validity was only anticipated for a small fraction of Phanerozoic time. The timespan for which agreement existed between data on this assembly was then taken as an estimate of the time validity of the assembly, and the APW path for that time interval was taken as a measure of the motion of the assembly relative to the pole. Additionally, however, discrepancies in older data between various source regions within the assembly threw light on preexisting continental components and how they converged; and divergences of younger data were indicative of the transposition from the re-assembly toward present day continent–ocean geography.

This approach has been more fruitful in the Precambrian: data from regions which are known to have been a tectonic entity at some *later* time (though not necessarily in the Precambrian itself) have been superposed in order to assess the validity or otherwise of the "working base-map". Commonly, the present-day continents have been chosen as tectonic entities of definite validity (even though that validity is *now* rather than at the geological time of interest). On this basis, APW paths for the Proterozoic of North America and Africa were presented by Irving and Park

(1972) and Piper et al. (1973) respectively. One problem of presentation that arises is the length and apparent complexity of these paths, arising from the length of time involved (ca. 2000 My) compared with the ca. 500 My to which readers are accustomed in viewing Phanerozoic paths. To alleviate this, more recent authors have usually drawn APW paths in segments covering a few hundred million years at a time. Even though this procedure utilizes data from more than one Proterozoic tectonic block at a time, problems of inadequacy of data remain to the extent that interpolation between data still produces apparent differences which may have no geophysical basis. McElhinny and McWilliams (1977) have given a reminder that in this context the proper procedure for comparing data sets is to test for identity rather than difference; only if the test of identity fails to gain adequate statistical support, can difference be hypothesised with any substantial degree of confidence. First, then, we will look at the extent of mutual compatibility of the data on present-day base maps (i.e. the extent to which the data are consistent with intracontinental orogenic belts forming between the cratons in each continent) and then turn to the differences (i.e. the extent to which the data indicate separate elements of continental crust in relative motion with each other).

3.2. Intracontinental orogeny

Piper et al. (1973) first pointed out the apparent compatibility of data as old as ca. 2300 My from various cratons in Africa (and South America, too, in its supposed Mesozoic position adjacent to Africa) with their present relative positions, despite the existence between those cratons of ca. 1850, 1100 and 550 My orogenic belts. The North American path (Irving and Park, 1972) also contained data from more than one structural province of the Canadian shield falling on a single master APW path. The interpretation of Piper et al. (1973) and Briden (1973) that some of these orogenic belts developed intra-continentally attracted much criticism (e.g. Burke et al., 1976) mainly because of uncertainties inherent in the data as pointed out by the original authors. Nevertheless this conclusion seems to be borne out by subsequent studies in different Proterozoic provinces in Canada (Irving and McGlynn, 1976a), within Australia (McElhinny and McWilliams, 1977; Embleton, 1978) and Africa (McWilliams et al., 1978). Piper (1976a, b) has taken this argument to its ultimate conclusion, contending that data in the age range 2700–1000 My from the major continental blocks are in general compatible with a single super-continent, reminiscent of early Mesozoic Pangaea, but with the familiar Gondwanaland and Laurasia displaced from each other so that southwestern U.S.A. abuts against Arabia. This extreme interpretation has attractions of geometrical simplicity. However the analysis on which it is based relies heavily on the assumption that uncertainties in relative ages to be assigned to many poles are great, and that many current assignments are wrong. It appears to be contradicted by the most intensive groups of data from Laurentia, Africa and Australia (e.g. compare Figs. 1a, c and d).

3.3. Small separations, closures and local rotations

As has been carefully pointed out by Piper et al. (1973) and all subsequent authors who have described compatibility of data from adjacent cratons, the uncertainties in palaeomagnetic pole positions and in the ages to be assigned to them are such that it is not possible to rule out the opening of small oceans (ca. 1500 km wide) which subsequently closed to the same relative positions from which they started. While such behaviour is not utterly incompatible with plate tectonic theory, nor with plate behaviour as it appears to have operated in Phanerozoic times, it is nonetheless a very special case and not what is generally expected; in general, motions of plates are assumed to be constrained only by their relationship to other extant plates and plate margins, and not by kinematic events several hundred million years before. McWilliams (1977) describes a case where small, but not subsequently self-cancelled, movements may have been detected. He contends that Proterozoic data from three cratons within Australia which now have late Proterozoic–early Palaeozoic sedimentary basins more than 1000 km wide between them are slightly more compatible if the cratons had originally been close-packed together and rotated to their present positions in the early Palaeozoic in a way already suggested by Harrington et al. (1973).

3.4. Large-scale relative motions

The principal evidence for large motions of individual cratons, or aggregates of cratons, relative to others lies in the differences in length and shape between those segments of Proterozoic APW paths which now appear to be reliably defined (Fig. 1). Thus the African path for the period around 2300–1850 My does seem to be genuinely different from the Laurentian and Australian paths for this interval. The Australian and Laurentian paths from about 1750 to 1200 My also appear to be significantly different. One problem in comparing Precambrian APW paths is the ambiguity of geographic polarity to be assigned to the paths. In Section 5 we discuss the possibility that distinctive polarity–time patterns in Precambrian APW paths might in some cases allow at least partial solution to this problem, and we describe an example where, using this approach for the Laurentian and African paths for the interval ca. 2000–1850 My, it is difficult to avoid the conclusion that these two continents were in relative motion at that time.

There remains a disturbing feature about the grouping of Precambrian cratons which apparently have common Proterozoic APW paths. It is that the grouping within North America, within Africa and within Australia has the common feature of grouping *within* present-day continents and the differences are *between* present-day continents. Yet although present-day continents have a present-day tectonic significance, we know that they were not distinctive units a mere 200 My ago, and hence the possibility that present-day continents have a Proterozoic tectonic significance is remote. It is, however, probably premature to worry about this possible paradox, because the data are still so sparse. It has already been argued (Piper et al., 1973) that the Brazilian craton was part of the African group (which was therefore not exclusively African). When regional coverage is improved in other continents other exceptions may emerge and the correlation may disappear.

4. Palaeomagnetism in orogenic belts

The foregoing discussion illustrates the difficulty of reaching unambiguous conclusions about global-scale Proterozoic tectonics on palaeomagnetic grounds alone, even when uncertainties in the meaning of the individual data are minimized by concentrating on unmetamorphosed and undeformed rocks on stable cratons. It might, then, be thought futile to compound the complexities by attempting to interpret data from orogenic belts in which the effects of one or more episodes of metamorphism, deformation, chemical alteration, burial and uplift must all be considered a priori. Perhaps this is why the early work of Balsley and Buddington (1957) was not promptly followed up by other workers. It would indeed have been foolhardy to do so without the degree of sophistication of alternating field, chemical and notably thermal demagnetization that has now been attained.

The Precambrian orogenic belt that has been most intensively studied palaeomagnetically is the Grenville belt in Canada. Studies on this belt, which began with that of Hargraves and Burt (1967), have been discussed and reviewed by Irving et al. (1974), Buchan and Dunlop (1976) and McWilliams and Dunlop (1978). The palaeomagnetism of the Grenville belt is discussed further below. Park (1973, 1975, 1977) has studied the palaeomagnetism of various rock units in the Canadian shield that were metamorphosed and folded in the Hudsonian orogeny, and Morgan (1976) and Beckmann and Mitchell (1976) have reported palaeomagnetic results from the metamorphic rocks of the Nagssugtoqidian belt in western Greenland. In Europe the palaeomagnetism of the regionally metamorphosed rocks of the Sveco–Norwegian zone of the Baltic shield has been investigated by Poorter (1972, 1975). Many of these studies revealed multi-component magnetizations, and the key to the analysis of these is progressive demagnetization together with vector subtraction to dissect remanence components according to their critical blocking temperature, coercivity or acid-solubility spectra. For success, each component must have part of its stability spectrum non-overlapping with any other component in at least part of the collection. It inevitably follows that larger collections and more laboratory tests and measurements are required, to resolve problems which are common in unmodified igneous or sedimentary rocks.

The majority of the magnetizations found in the previously deeply buried rocks of orogenic belts are considered to have been acquired during post-

orogenic uplift and cooling. When multi-component magnetizations are present they are often given an age sequence based on the simple argument that components with lower blocking temperatures (or coercivities) are younger than components with higher blocking temperatures or coercivities. Whilst this is probably often true it is important to bear in mind that superimposed CRMs can have any blocking temperature or coercivity, and that for these components relative stability is no clue to relative age. Many unmetamorphosed or very weakly metamorphosed rocks from the Canadian shield have been shown to have both a primary magnetization dating from their time of formation and also one or more secondary overprints (Evans and Bingham, 1976; Irving and McGlynn, 1976b; McGlynn and Irving, 1978). These overprints are thought to have been acquired during uplift following deep burial and mild reheating, illustrating that even rocks from relatively stable areas can carry a complex magnetic record.

It appears that anisotropy due to metamorphic fabric is not as serious a distorting influence on the direction of remanence as was once thought. McElhinny (1973) has given simple theoretical arguments why this should be so, and Park (1973, 1977) and Irving and Park (1973) have produced supporting experimental evidence, although Beckmann et al. (1977) have proposed an empirical method, which appears to have no valid basis, of using low field anisotropy to correct remanence directions.

The palaeomagnetic record in Precambrian orogenic belts is strongly influenced by the long slow post-orogenic cooling that these terrains experience, cooling rates of a few degrees per My over periods of ca. 10^8 y probably not being uncommon. This slow cooling is frequently reflected in the record contained within different lithologies of the same rock unit, and often within individual specimens, when remanence carriers with successively lower peaks of blocking temperatures have distinctly different magnetizations which define successively younger points on the APW path. The slow cooling is also reflected in the regional pattern of palaeomagnetic results when various parts of the currently exposed erosion level through a belt cooled at different times – the palaeomagnetic results from the various parts of the belt again defining various points on the appropriate segment of the APW path. This situation is well-illustrated by the palaeomagnetic results from the Grenville belt, recently discussed by McWilliams and Dunlop (1978). Most of the Grenville belt has experienced amphibolite or granulite facies metamorphism, and as the experimental and theoretical work of Pullaiah et al. (1975) suggests that little, if any, remanence can survive high greenschist facies metamorphism, it is unlikely that any remanence could have survived the Grenville orogeny, and so the magnetizations recorded in Grenville rocks were almost certainly acquired during post-orogenic uplift and cooling. Harper (1967) and Baer (1976) produced maps showing the cooling history of the Grenville belt in terms of equal-age thermochrons based on K–Ar mica ages. McWilliams and Dunlop have shown that if the palaeomagnetic results from the various parts of the belt are grouped according to the position of the sampling areas relative to those thermochrons, then when the resulting poles are plotted in time sequence they define a section of the APW path which has the form of a simple loop. Further, many studies in the Grenville belt have revealed multi-component magnetizations, both within different rock types in the same localised study and also residing together in individual specimens. These magnetizations fall on the APW path loop in a way which is generally consistent with acquisition of the higher blocking temperature magnetizations before the lower blocking temperature magnetizations.

It should be mentioned that some earlier interpretations of the data from the Grenville belt have been in terms of the collision of a previously separated Grenville plate with the rest of the Laurentian Shield at the time of the orogeny, because the APW path from Grenville rocks appeared to be distinct from the APW path for the rest of the shield, and converged on the main path by the end of the orogeny. However, there is much evidence to suggest that the Grenville orogeny and hence subsequent uplift and cooling occurred while the Grenville belt was in its present position relative to the Superior province. Because all the Grenville magnetizations recognized at present appear to be post-orogenic cooling magnetizations, they were therefore acquired with the Grenville belt in its present relative position and so the Grenville APW path loop would appear to be a detailed modification of the Laurentian Shield APW path and not indicative of independent movement

and collision. The Grenville palaeomagnetic record can therefore provide no evidence about possible movements of the belt before and during the Grenville orogeny.

Several aspects of the palaeomagnetism of slowly cooled high-grade metamorphic terrains were discussed by Morgan (1976). It was pointed out that because of very slow cooling even rocks with only one type of remanence carrier and a single fairly narrow range of blocking temperatures should nevertheless contain the magnetic record of an appreciable period of time. If cooling through the range of blocking temperatures occurred during a period of sufficiently fast APW then a short but continuous segment of the APW path should be recorded within individual specimens and be retrievable by vector analysis of progressive demagnetization results. Also, because of the long magnetization period, palaeosecular variation should be averaged out within each *specimen*, so that both "within-site" and "between-site" scatter due to this cause should be absent. Thus the mean pole for each sampling site should be a time-averaged palaeomagnetic pole and not a spot-reading virtual pole. It would be expected however that there would be scatter of site mean poles due to APW if the various sites in a study have different mean blocking temperatures, or if the sites are distributed through different structural levels in the terrain. This scatter should of course also define the same APW path segment as revealed by progressive demagnetization of individual specimens. The structural level is important because cooling magnetizations are acquired as a given level in the crust passes up through the isotherms corresponding to the range of magnetic blocking temperatures, and so it would be expected that, with horizontal isotherms, upper levels would have the oldest magnetizations and deeper levels would have successively younger magnetizations.

Morgan (1976) described results from a high-grade terrain at the southern margin of the Proterozoic Nagssugtoqidian mobile belt in western Greenland which appeared to show all the features just described. The site poles were arranged in a narrow linear trend and there was some evidence that this was due to the sites being at different structural levels. Also, progressive demagnetization at many sites (which incidentally had only a single peak of blocking temperatures between about 540 and 580°C) caused the magnetization to move systematically in a direction which was almost exactly along the trend of the site poles. When these results are corrected for the drift of Greenland away from Canada, the segment of the APW path so defined falls on and almost parallel with the latest version of the Laurentian APW path (McGlynn and Irving, 1978) for the period approximately 1700 to 1600 My, and the sense of time along the segment as deduced from the movements during demagnetization is in accord with the ages assigned to the published path. This section of the Laurentian APW path is characterized by many secondary magnetizations that were acquired during slow uplift and cooling following the Hudsonian orogeny.

Because of the very long time-periods represented by the magnetizations in a slowly cooled rock it might be expected that the resultant would be self-cancelling due to repeated reversals of the geomagnetic field during the magnetization interval (Beckmann, 1976). However, the fact that there have now been many studies of slowly cooled rocks which have revealed coherent magnetizations suggests that there were periods of the order of 10^7–10^8 y either of constant polarity or of biased reversals with one polarity persistently dominant. Thus in slowly cooled rocks, reversals within component magnetizations should be rare or absent, and it is perhaps significant that of the 23 component magnetizations listed by McWilliams and Dunlop (1978) for the slowly cooled Grenville belt only four contain mixed polarities, and in each of these one polarity is strongly dominant. It is noted that the study of polarity bias in the Phanerozoic by Irving and Pullaiah (1976) indicated that even polarity balance is comparatively rare, and that the Phanerozoic appears to have been characterized by four long intervals with alternately normal and reverse bias. If geomagnetic field behaviour was similar in this respect in the Precambrian then slowly cooling rocks would have had a high probability of acquiring non-cancelling magnetizations.

Palaeomagnetic investigations of metamorphic terrains in orogenic belts can therefore, in favourable circumstances, provide unusually detailed information about certain segments of APW paths. They also have other potential uses, for example providing estimates of cooling and erosion rates and geothermal gradients, although these estimates will inevitably suf-

fer from uncertainties as to whether the magnetic record in a belt reflects a single period of very slow cooling or an earlier cooling followed by one or more milder reheating events. There will also be uncertainties in relating blocking temperatures observed in the laboratory to the effective blocking temperatures during slow cooling (Pullaiah et al., 1975; Dodson and McClelland-Brown, 1980). In principle, however, if well-defined and well-dated APW paths are already known, then the presence of various magnetizations with different blocking temperatures should allow the cooling rate to be determined. Further, if palaeomagnetic results from known different structural levels are obtained, and these fall on different parts of the APW path segment according to depth in the way described above, then this could be used to determine the rate of erosion. Because cooling rate, erosion rate and geothermal gradient are simply related (cooling rate = erosion rate × geothermal gradient), then we should in principle be able to determine the geothermal gradient. Palaeomagnetic studies can also, in certain circumstances, provide information about the duration and intensity of thermal/metamorphic events, and in belts with complex diachronous cooling histories it is possible that palaeomagnetism may provide a more precise record of this history than can be obtained from radiometric studies.

5. New palaeomagnetic results from the Kaapvaal craton and Limpopo mobile belt, and the African APW path ca. 2000–1850 My

Palaeomagnetic results from Africa for the interval ca. 2000–1850 My come largely from the Kaapvaal and Rhodesian cratons and the intervening Limpopo mobile belt in southern Africa. In this section we describe work currently in progress on the Kaapvaal craton and Limpopo belt, which although as yet incomplete illustrates many of the general points made in the previous sections, and in the case of the Limpopo belt typifies the results that can be expected in a large-scale regional study of a metamorphic terrain.

The African APW path for the interval ca. 2000–1850 My is illustrated in Fig. 7, which shows all the previously published data and the new results reported here. The transit of the African APW path from east to west across North Africa during this time interval was first pointed out by Jones and McElhinny (1967) on the basis of data from the Bushveld Gabbro (1950 ± 50 My) from the Kaapvaal craton and the Mashonaland Dolerites (now dated at 1850–1910 My by Compston and McElhinny, 1975) from the Rhodesian craton, which delineate the same polar shift as is seen through the stratigraphic sequence of the lower part of the Waterberg Supergroup in the northern Transvaal. Other subsequently published results which reinforce this part of the APW path are studies by Hargraves (1970) on the ca. 1970 My Vredefort structure and Losberg Gabbro in the Transvaal, by Sabaté and Lomax (1975) on various intrusions (Aftout Gabbro, Yetti Dykes and Yetti Eglab Quartz Diorite) from the north-west Sahara dated at very approximately 1900 My, and by Jones et al. (1975) on the extensions of the Great Dyke into the Limpopo belt, which are thought to have been magnetized at about 1935 My (see below).

5.1. New data from the Kaapvaal craton

5.1.1. Rust der Winter

Although the original palaeomagnetic correlation of the age of the Waterberg sediments with the Bushveld Gabbro and Mashonaland Dolerites was attractive and compatible with the evidence, the interpretation would have been more secure if the relationship of the palaeomagnetically sampled sediments to an acid volcanic horizon at Rust der Winter, dated at 1790 ± 70 My by Oosthuyzen (1964), had been known; unfortunately it was uncertain whether that volcanic rock was intrusive or extrusive, and whether the sediments with which it is interbedded are to be correlated with the lower Waterberg. The uncertainty in the minimum age to be attached to the lower Waterberg sediments has now been eased, though not removed, by palaeomagnetic sampling at Rust der Winter. The red sediments immediately underlying the acid volcanic layer have a stable remanence isolated by AF and thermal cleaning as illustrated in Fig. 2. The pole corresponding to the site mean direction falls on the APW path approximately mid-way between the two oldest Waterberg poles W1 and W2 (Fig. 7), so the correlation of the Rust der Winter sediments with the lower Waterberg appears to be

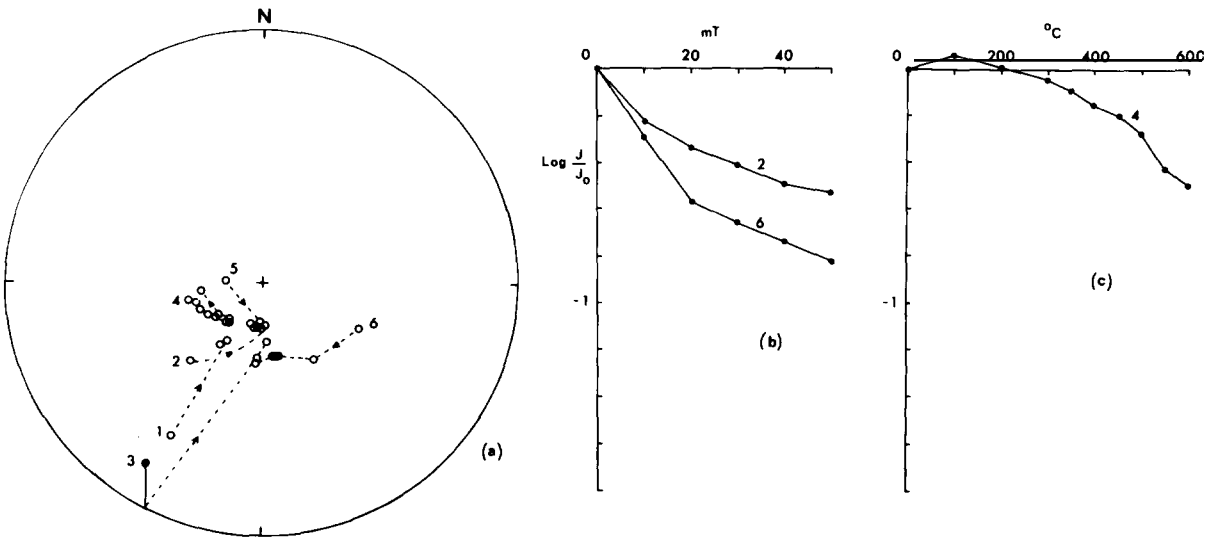


Fig. 2. Palaeomagnetic results from sediments underlying the volcanic horizon at Rust der Winter. (a) Changes of direction. (b) and (c) changes of intensity of remanence during progressive AF (specimens 1–3, 5 and 6) and thermal (specimen 4) demagnetization. (a) is a stereographic projection, solid circles lower hemisphere, open circles upper hemisphere. Results from the five AF treated specimens at fields of between 10 and 30 mT and the thermal result at 450°C were combined to yield a mean direction with $D = 190°, I = -66°, k = 64, \alpha_{95} = 8.4°$ and a corresponding palaeomagnetic pole at 16°S, 215°E.

safe. The polarity of the magnetization however is opposite to that of all the other magnetizations in that part of the APW path. It is noted that the Waterberg pole B of McElhinny (1968) is of the same polarity as the Rust der Winter pole, and also falls in the same part of the APW path (close to the pole YD). McElhinny however considers that the sediments from which these results were obtained are stratigraphically higher than those yielding pole W3 in Fig. 7, and so must be on a younger return loop of the path.

The remanence in the volcanic layer is complex and dominated by random strong soft components. At none of three sites does AF demagnetization yield significant mean directions and it is unlikely that a stable remanence has been isolated. The only possibility to the contrary is suggested by the fact that "remagnetization circles" of several specimens from all sites came close to intersection near $D = 340°, I = -60°$. This is quite different from the stable remanence in the adjacent sediments, but is almost identical to the direction of the present Earth's field in the sampling area, suggesting that this component is of very recent origin. Thus despite the fact that the volcanic horizon is now regarded by geologists of the South African Geological Survey as an extrusive felsite, this conclusion finds no palaeomagnetic corroboration and is still unsafe. However, the 1790 ± 70 My age on the felsite can now be regarded as a minimum age not only for its host sediments but also for its palaeomagnetically correlated Lower Waterberg equivalents, and hence there are no palaeomagnetic or age inconsistencies between the data from the Waterberg sediments and the various igneous bodies studied.

5.1.2. The Palabora igneous complex and related intrusions

The Palabora igneous complex in northeastern Transvaal is a carbonatite–pyroxenite–fenite central complex ca. 2000 My old associated with a large number of syenite bosses and cut by northeast–southwest trending dolerite dykes (Fig. 3). Briden (1976) reported results from 11 sites in which two groups of stable remanence directions were found. The first, largely derived from pyroxenites which form one of the oldest phases of the central intrusive complex, corresponded to pole positions close to that of the Bushveld Gabbro, while the second group,

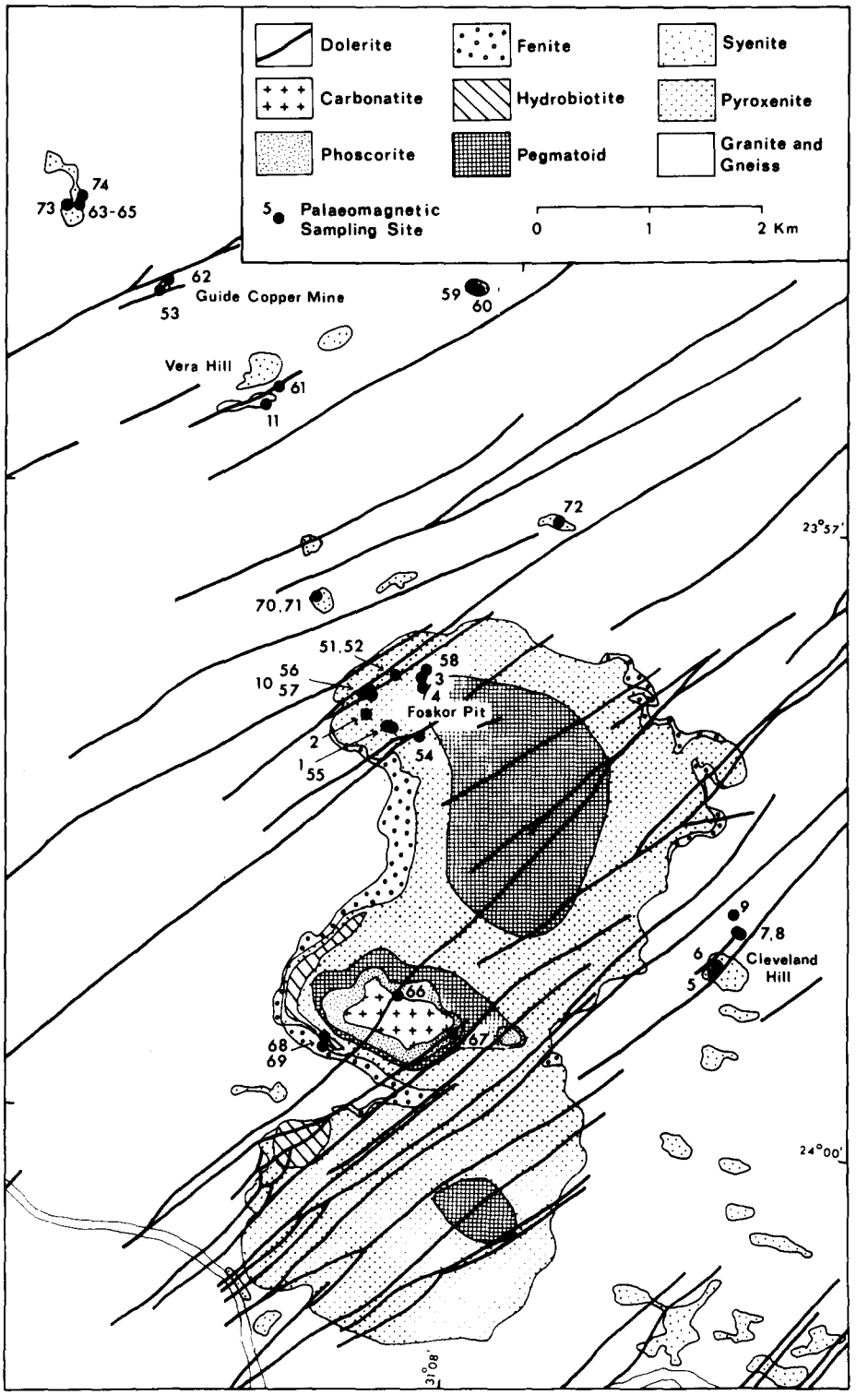


Fig. 3. Simplified geological map of the Palabora igneous complex (after Hanekom et al., 1965) showing the palaeomagnetic sampling sites.

from the separate Cleveland Hill Syenite plus a dolerite dyke which cut it, corresponded to pole positions close to the Mashonaland Dolerites. The lithological distinction among the palaeomagnetic groups were blurred, though, by the fact that another outlying syenite at Vera Hill was magnetized with the older group, as was a single site (No. 6) in the Cleveland Hill Syenite. Apart from this complication, the initial study was of interest because the two distinct magnetization directions indicated two apparent bursts of igneous activity in the same area ca. 100 My apart, and because the dyke [which had been surmised to be of Karroo (ca. 200 My) age] was evidently part of the Proterozoic activity. This latter point was confirmed by a single K–Ar whole-rock age determination of ca. 1520 My on the palaeomagnetic sample itself by R.M. McIntyre (pers. communication, 1975) which is to be regarded as a minimum estimate.

A further study of 24 more sites, of which 13 were, like all 11 original sites, in artificial exposures (working mines and associated excavations, and old quarries), has now been carried out. Except for three which showed instability, all sites in artificial exposures yielded coherent palaeomagnetic data. The

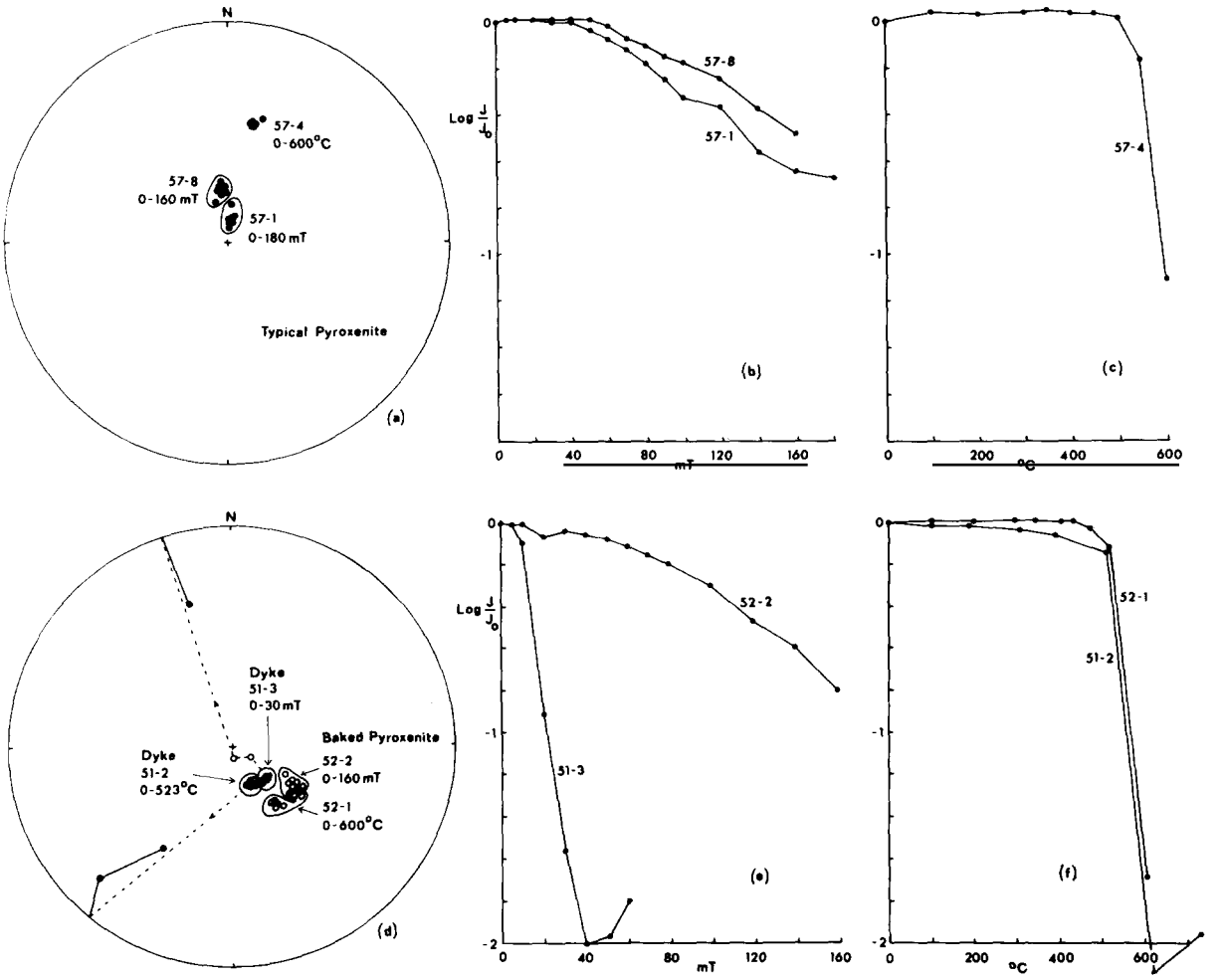


Fig. 4. Palaeomagnetic results from the Palabora igneous complex. Changes of direction and intensity of remanence during progressive thermal and AF demagnetization of (a)–(c) a typical pyroxenite, and (d)–(f) a typical dyke and a pyroxenite baked by that dyke. (a) and (d) are stereographic projections, filled circles lower hemisphere, open circles upper hemisphere.

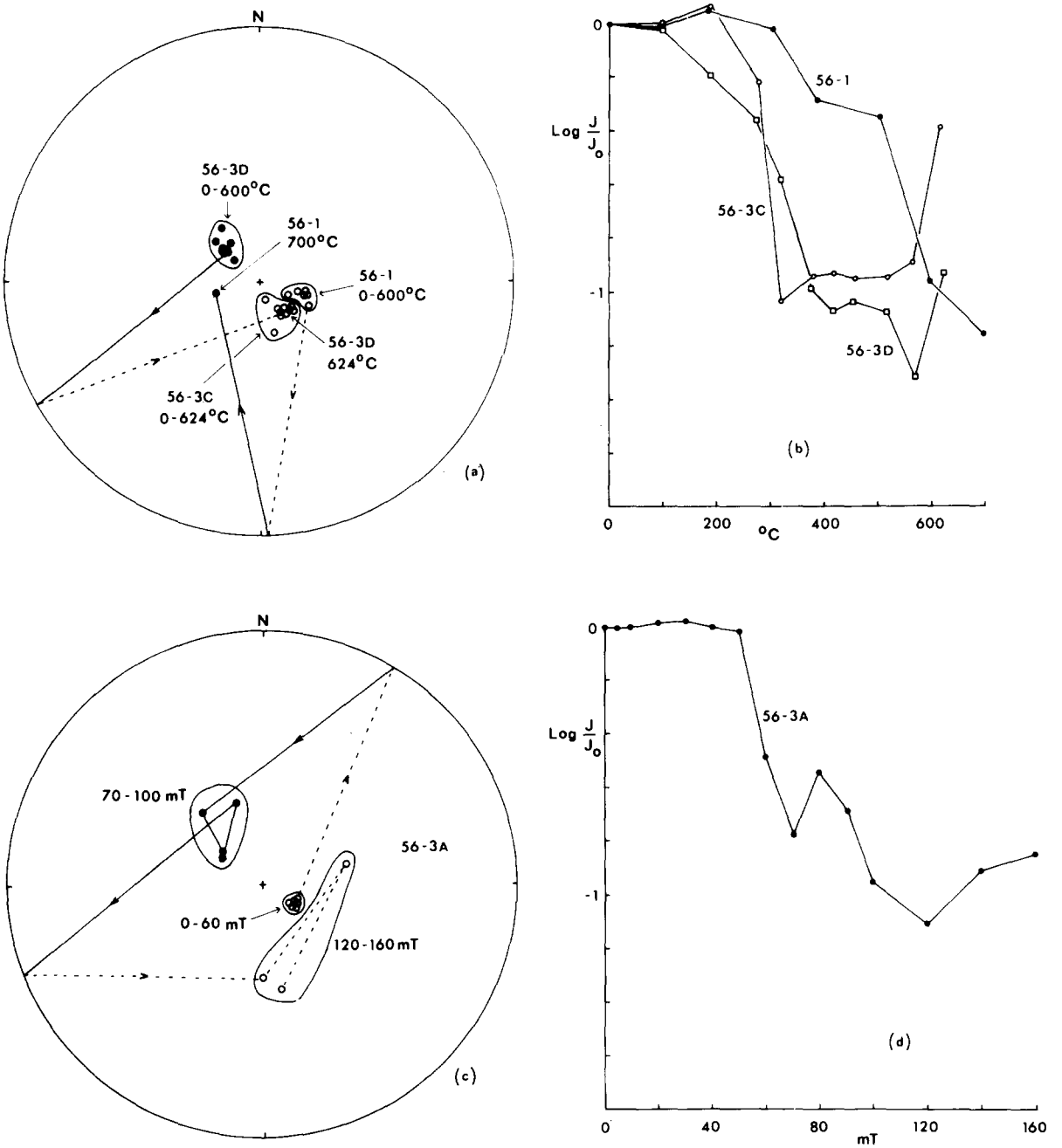


Fig. 5. Palaeomagnetic results from dyke at site 56 at Palabora showing reversals during (a) and (b) progressive thermal, and (c) and (d) progressive AF demagnetization. (a) and (c) are stereographic projections of directions, filled circles lower hemisphere, open circles upper hemisphere.

other 11 sites were surface outcrops of outlying syenites and dykes cutting them, and none yielded a systematic magnetization. They all have extremely strong and magnetically soft NRMs, undoubtedly of IRM origin, with coercivities reaching sufficiently high to prevent isolation of any underlying stable component by AF demagnetization; they are not considered further. Deprived of evidence about these outlying bodies, the new study concentrated on the central complex itself and dykes cutting it, an outlying pyroxenite at the Guide Copper Mine, and a dolerite dyke (in a small disused quarry) which cut the Vera Hill syenite referred to above.

Representative AF and thermal demagnetization characteristics of the newly reported sites are shown in Figs. 4 and 5. The majority of sites have single component NRM uncontaminated by secondary components of any kind. The NRM directions of the pyroxenites in the Foskor mine are particularly consistent, and are very stable to both AF and thermal demagnetization (Fig. 4a, b, c). The thermal demagnetization curves show that the remanence is carried by grains with blocking temperatures between 500 and 600°C, and remanence-coercivities of generally more than 50 mT. The only exceptional direction was at site 52, within 20 cm of the contact of the dyke which was sampled at sites 51 and 56, where the pyroxenite is completely remagnetized into the dyke direction (Fig. 4d, e, f). Some of the samples from shallow quarries had secondary components which were removed by AF or thermal cleaning. At site 56, progressive AF and thermal demagnetization of some specimens indicated the presence of both normal and reversed components within the same specimens (Fig. 5). The majority of specimens, e.g. 3C, had steep negative inclinations throughout the demagnetization range, but one (3D) had the opposite direction up to 600°C and then showed a perfect reversal into the more usual direction after demagnetization at 624°C, while specimen 1 had the usual direction up to 600°C but showed a reversal into the exceptional direction after heating to 700°C. Progressive AF demagnetization of specimen 3A resulted in two reversals; up to 60 mT the usual negative inclination was seen, but after 70 mT there was a reversal into the opposite direction which was maintained up to 100 mT, followed by a reversal back to the original direction at fields of 120 mT and above. The interpretation of these reversals in individual specimens poses a difficult problem. Reheating effects due to multiple intrusions might be one possible explanation, but there was no field evidence to support this. Alternatively the reversed components might be due to a superimposed later CRM. The near antiparallelism of the directions suggests that the later components were acquired soon after the original magnetizations, but of course after sufficient time for a reversal to be recorded.

Data from all 21 sites yielding coherent results are summarized in Table I and the site mean directions are shown in Fig. 6. The division of the remanence directions into two groups indicated by the original study is confirmed, and it is now seen that whereas the Group 1 magnetizations are all of the same polarity (with northerly positive directions), the Group 2 magnetizations are of mixed polarity. Reference to the APW path in Fig. 7, where the polarity of the constituent poles is indicated, shows that the polarity pattern seen in the Palabora results is the same as that shown by the APW path as a whole, i.e. almost exclu-

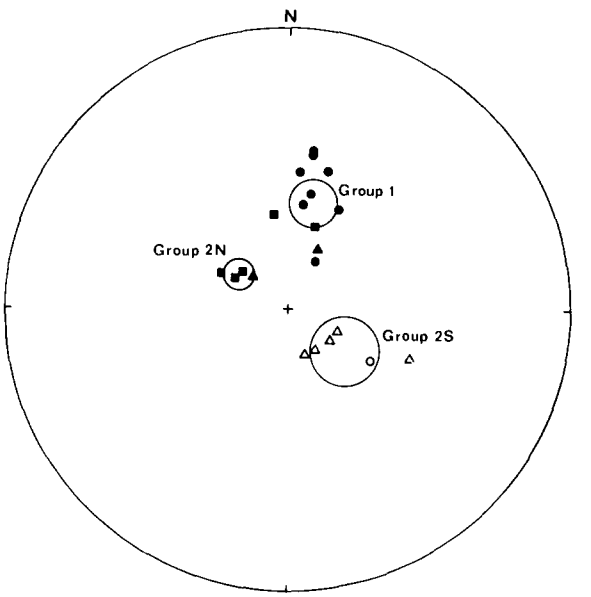


Fig. 6. AF and thermally cleaned site mean directions from the Palabora igneous complex. Circles, pyroxenites; triangles, dolerite dykes; squares, syenites. The circles of 95% confidence about the three group mean directions (Table I) are also shown. Stereographic projection, filled symbols lower hemisphere, open symbols upper hemisphere.

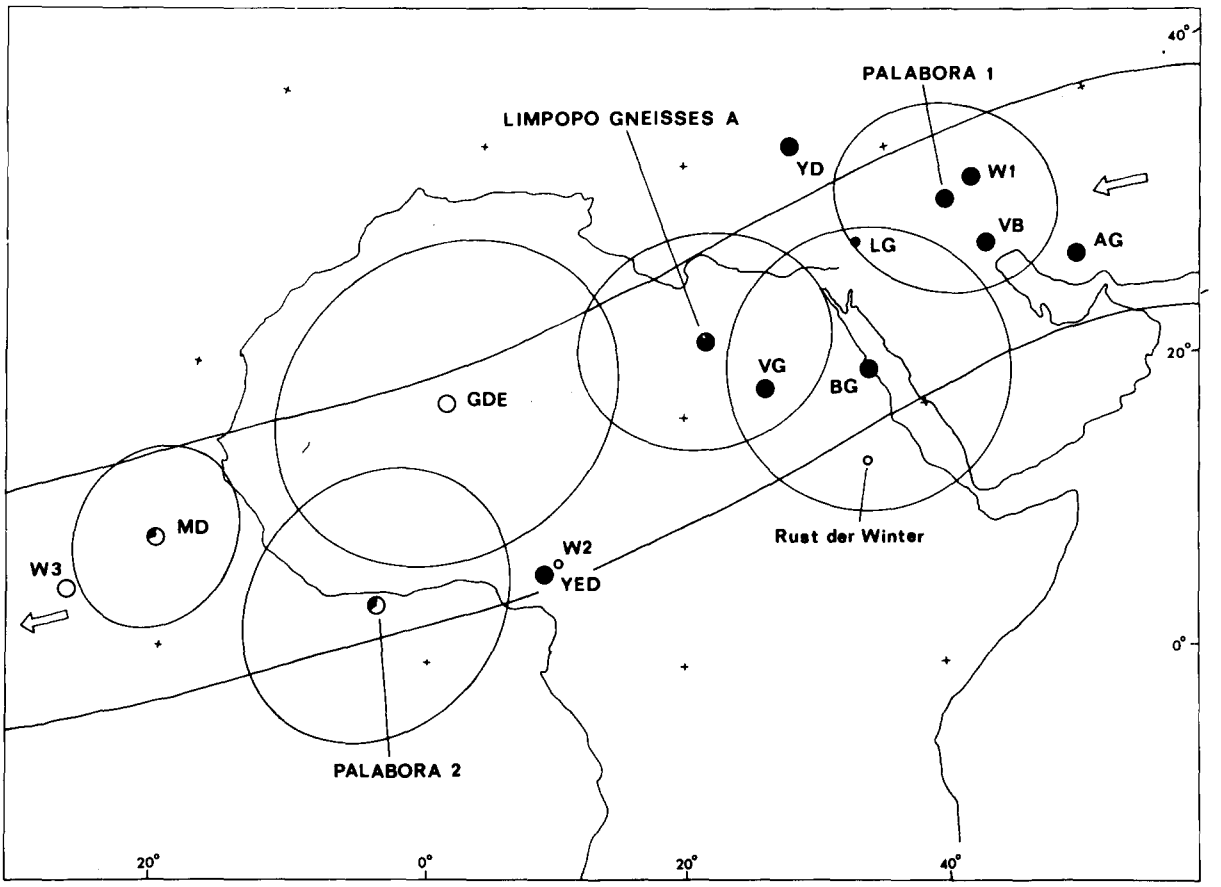


Fig. 7. APW path for Africa for the interval ca. 2000 to 1850 My. Previously published results are shown by abbreviations as follows: BG, Bushveld Gabbro (Gough and Van Niekerk, 1959); GDE, Great Dyke extensions into Limpopo belt (Jones et al., 1975); MD, Mashonaland Dolerites (McElhinny and Opdyke, 1964); W1 to W3, Waterberg sediments (stratigraphically arranged poles E, D and C respectively in McElhinny, 1968); LG, Losberg Gabbro; VB, Vredefort sites 9, 10 and 11; VG, Vredefort Granophyre (Hargraves, 1970); AG, Aftout Gabbro; YED, Yetti Eglab Quartz Diorite; YD, Yetti Dykes (Sabaté and Lomax, 1975). New results fully labelled. Filled circles, poles corresponding to N-seeking magnetizations; open circles, poles corresponding to S-seeking magnetizations; partly filled circles, both polarities in proportions indicated. Poles from major studies are shown with oval of confidence, small circles represent poles based on single sites only.

sively northerly magnetizations in the older part of the path containing the Palabora 1 pole, and mixed but predominantly southerly magnetizations in the younger part of the path containing the Palabora 2 pole.

The distribution of rock types among the two directional groups confirms that neither the syenites nor the dykes are all of a single age. It also confirms the close association of dyke emplacement with that of the syenites and pyroxenites. The Proterozoic rather than Karroo age of these dykes has now been further confirmed by two single whole-rock K–Ar age determinations (D.C. Rex, pers. communication, 1978) on the palaeomagnetic samples of site 66 (which belongs to Group 2S) of 1992 My and site 67 (which is palaeomagnetically unstable and therefore not listed in Table I) of 1575 My, the latter age being regarded as a minimum.

The published data and that just described show the widespread extent of igneous activity at ca. 2000–1850 My in the Rhodesian and Kaapvaal cratons. Apart from the Bushveld and neighbouring

TABLE I

Palaeomagnetic results from the Palabora igneous complex

Site		Rock type	Treatment							
			mT	°C	N	R	k	α_{95}	D	I
Group 1 *1										
1	Foskor Pit	pyroxenite	60	–	5	4.93	60	10.0	5	+38
2			60	–	5	4.89	37	12.7	16	+36
3			60	–	7	6.80	31	11.1	9	+31
4			60	–	5	4.94	64	9.6	27	+47
10			50–80	–	6	5.93	77	7.7	9	+32
57			60	400	8	7.70	23	11.7	11	+45
55		diabase	30	400	6	5.90	50	9.5	8	+49
53	Guide Copper Mine	pyroxenite	20	200	4	3.96	67	11.2	30	+68
11	Vera Hill	syenite	40–60	–	3	2.89	18	29.8	19	+56
61	Vera Hill	dolerite dyke	40	400	8	7.87	55	7.5	27	+64
6	Cleveland Hill	syenite	40–60	–	5	4.93	60	10.0	352	+53
Group 2N *2										
5	Cleveland Hill	syenite	50–60	–	5	4.85	26	15.3	301	+66
7		syenite	50–60	–	5	4.98	268	4.7	299	+60
9		syenite	50–60	–	4	4.00	751	3.4	310	+67
8		dolerite dyke	60–80	–	4	3.98	175	7.0	313	+71
Group 2S *3										
51	Foskor Pit	dolerite dyke	30	400	8	7.97	215	3.7	146	–70
52		baked pegmatitic pyroxenite at dyke contact	60	500	3	2.93	31	22.3	122	–52
54		dolerite dyke	40	400	6	5.89	47	9.8	159	–70
56			60	600	6	5.94	81	7.4	126	–69
58			40	400	6	5.96	122	6.0	113	–68
66	PMC Pit		–	400	8	7.93	105	5.4	112	–40

*1 Site analysis: $N = 11, R = 10.694, k = 33, \alpha_{95} = 8.1, D = 13, I = +48$. Palaeomagnetic pole = 35°N, 45°E ($dp = 7°$, $dm = 11°$).

*2 Site analysis: $N = 4, R = 3.987, k = 234, \alpha_{95} = 6.0, D = 305, I = +66$. Palaeomagnetic pole = 2.5°N, 358°E ($dp = 8°$, $dm = 10°$).

*3 Site analysis: $N = 6, R = 5.828, k = 29, \alpha_{95} = 12.6, D = 126, I = -62$. Palaeomagnetic pole = 6.4°S, 175°E ($dp = 15°$, $dm = 20°$).

Groups 2N + 2S normalized: Site analysis: $N = 10, R = 9.811, k = 47, \alpha_{95} = 7.0, D = 305, I = +64$. Palaeomagnetic pole = 4°N, 356°E ($dp = 9°$, $dm = 11°$).

intrusions, and the Palabora central complex, it now emerges that dolorite dykes at Palabora (and by implication over a wide area of the Transvaal) and the Mashonaland Dolerites also date from this period. It is noteworthy that the age of both these suites of dolerites was first realized from palaeomagnetic evidence, only as a consequence of which were radiometric age determinations undertaken.

5.2. New data from the Limpopo mobile belt

The Limpopo belt is a linear zone of strongly deformed high-grade metamorphic rocks situated between the Archaean Rhodesian and Kaapvaal cratons (Fig. 8). The predominant lithologies are tonalitic and granitic gneisses, but there are subordinate amounts of supracrustal rocks which have been

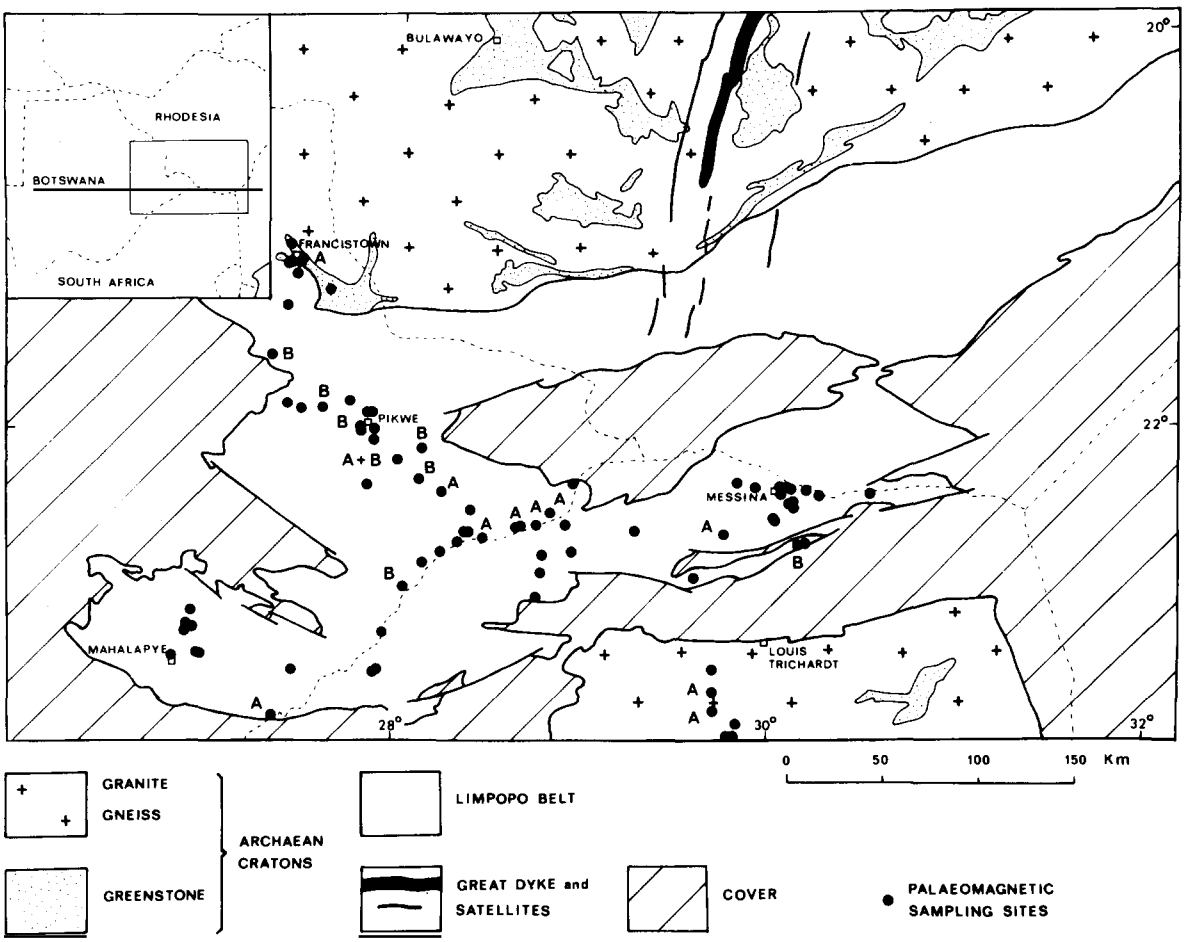


Fig. 8. The Limpopo mobile belt and adjoining parts of the Rhodesian and Kaapvaal cratons. Sampling sites where A and B magnetizations have been found are indicated.

infolded and suffered amphibolite and granulite facies metamorphism together with the basement. The belt has been extensively studied geochronologically. A large number of Rb–Sr whole-rock isochrons have been determined from various parts of the belt, and although the total range of these ages is from 3860 to 1820 My, the great majority fall in the range 2800–2500 My and are thought to date the peak of the last major episode of metamorphism and deformation in the belt (Van Breemen and Dodson, 1972; Hickman and Wakefield, 1975; Barton and Ryan, 1977). More important from the palaeomagnetic point of view, however, is the fact that 23 Rb–Sr mineral ages from various localities in the belt show a very pronounced peak at 2000 My (Van Breemen and Dodson, 1972). Whether these 2000 My mineral ages represent uplift and cooling after a long and continuous period of deep burial, or a later thermal event distinct from an earlier cooling episode, is not known. What is important is that the whole area last cooled through the blocking temperatures for Rb–Sr mineral systems at about 2000 My, and the main objective of the palaeomagnetic study was to identify and investigate the magnetization acquired during this cooling event. No later events other than slight tilting, block faulting, deposition and erosion of thin cover, and the intrusion of minor dyke swarms are known, so the Limpopo belt offers an ideal situation for the palaeomag-

netic study of regional cooling free from obvious severe disturbing influences.

Throughout most of the belt there is no clear trend of mineral ages with geographical position, but in the northern margin of the belt in Rhodesia the ages appear to be generally slightly younger. The mean of five mineral ages from the northern margin (all in the general vicinity of the extensions of the Great Dyke and its satellites into the belt) is 1935 My, whereas the mean of the remaining 18 mineral ages (from Botswana, South Africa and the extreme south of Rhodesia) is 2020 My. There is thus an indication that the more northerly part of the belt in Rhodesia cooled approximately 100 My later than the rest of the belt, a fact which should be detectable palaeomagnetically. The only previous study in the Limpopo belt was by Jones et al. (1975) on the extensions of the Great Dyke and its satellites into the northern part of the belt in Rhodesia. These "extension" dykes have yielded a Rb–Sr whole-rock isochron age of 2600 My, similar to the age of the Great Dyke in the Rhodesian craton, but are apparently unaffected by Limpopo deformation or metamorphism. The mean palaeomagnetic pole for seven sites in these dykes, however, is significantly different from the Great Dyke pole and falls on the APW path between the poles for the Bushveld Gabbro and the Mashonaland Dolerites (Fig. 7), suggesting that the magnetizations in these dykes was set (or reset) in the ca. 2000 My thermal event in the belt. The evidence

TABLE II

Preliminary palaeomagnetic results from Limpopo belt gneisses

Site	Peak AF (mT)	N	R	k	α_{95}	D	I
Group A *1							
10	15	5	4.64	11.0	24.1	350.7	+57.1
15	140	5	4.98	248.3	4.8	4.9	+57.6
16 LCR	*3	11	10.54	21.6	10.0	175.3	−63.1
16 HCR	100	12	11.61	28.3	8.2	42.9	+58.1
22	NRM	6	5.84	31.8	12.0	334.5	+77.1
29 MCR	*4	7	6.95	124.8	5.4	351.4	+65.8
31	80	7	6.62	15.8	15.6	357.1	+57.1
32	90	6	5.99	1081.6	2.0	349.7	+54.3
33	120	7	6.88	48.5	8.7	318.8	+53.0
36	100	5	4.32	5.8	34.4	320.6	+44.7
38 MCR	*5	8	7.39	42.8	8.5	5.2	+48.2
48	NRM	6	5.99	589.4	2.7	0.9	+61.0
Group B *2							
11	NRM	7	6.78	27.2	11.7	53.9	− 7.3
25	20	6	5.64	13.7	18.7	55.9	+ 1.6
37	NRM	6	5.99	346.5	3.6	37.8	−27.3
38 LCR/HCR	*6	8	7.89	65.3	6.9	43.7	−29.1
39	NRM	9	8.68	24.9	10.5	227.0	+38.1
42	NRM	10	9.71	31.3	8.7	26.6	−23.7
45	NRM	7	6.90	62.3	7.7	33.7	+26.1
52	NRM	6	5.96	140.2	5.6	43.3	+20.1

*1 Site analysis with site 16 LCR direction reversed: $N = 12, R = 11.65, k = 31.6, \alpha_{95} = 7.8, D = 352.2, I = +59.9$; Palaeomagnetic pole = 26°N, 22°E ($dp = 9°$, $dm = 12°$).

*2 Site analysis with site 39 direction reversed: $N = 8, R = 7.30, k = 10.0, \alpha_{95} = 18.3, D = 43.1, I = -10.1$. Palaeomagnetic pole = 45°N, 102°E ($dp = 9°$, $dm = 18°$).

*3 Site mean computed from removed vectors at low fields, generally between 30 and 40 mT.

*4 Site mean computed from removed vectors between 120 and 180 mT.

*5 Site mean computed from removed vectors between 60 and 120 mT (see Fig. 11).

*6 Site mean computed from removed vectors between 5 and 20 mT and observed magnetizations at 180 and 200 mT (see Fig. 11).

of the mineral ages from the gneisses in that part of the belt discussed above would suggest that the magnetization was acquired at about 1935 My.

In the present study of the Limpopo belt, samples have been collected from over 80 sites in South Africa and Botswana (Fig. 8); the Rhodesian part of the belt has unfortunately been inaccessible for field sampling. A few sampling sites were also located in the northern part of the Kaapvaal craton. The majority of samples were taken from the predominant granitic gneisses, but a few sites were taken in metasediments and metavolcanics of the infolded supracrustal sequences. Work on this material, which is still in progress, has shown that of the sites so far examined many had widely scattered total NRM directions which did not improve on demagnetization. This may be due to superficial effects such as lightning and weathering. However, at 18 sites (mainly from relatively fresh low exposures in stream beds) there was good within-site grouping (Table II) either of the total NRM or after AF cleaning. These site mean directions fall into two groups (Fig. 9); there is a well-defined group with northerly declinations and positive inclinations (group A) and a less well-defined group with northeasterly declinations and moderate to shallow both positive and negative inclinations (group B). Each group contains one site which is reversed with respect to the remainder.

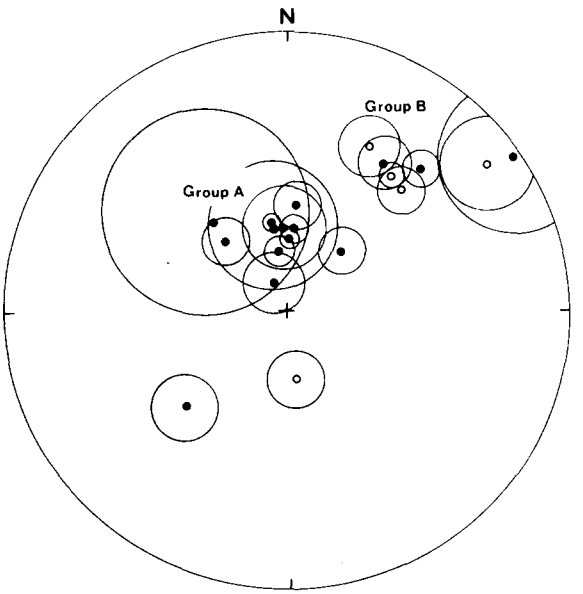


Fig. 9. AF cleaned site mean directions of magnetization from Limpopo Gneisses with circles of 95% confidence (see Table II). Stereographic projection, filled circles lower hemisphere, open circles upper hemisphere.

Let us first consider the A magnetizations. They are found at sites throughout the area of the belt investigated, and also at two sites in the northern part of the Kaapvaal craton. At some of these sites the directions were well-grouped at the NRM stage and the grouping did not improve with AF demagnetization. At most sites, however, the NRM directions were widely scattered and it was only after demagnetization to moderate or high fields that good grouping and stable end-points were obtained, and at three sites the A magnetization was best revealed by analysis of the vectors removed during part of the progressive demagnetization sequence. Preliminary thermal demagnetization indicates that the A magnetization resides in grains with a limited critical blocking temperature range between about 550 and 600°C.

The palaeomagnetic pole corresponding to the mean A magnetization falls squarely on the African APW path close to the poles for the Bushveld Gabbro and Vredefort Granophyre, i.e. at a point corresponding to an age of about 2000–1950 My (Fig. 7). Thus it seems very probable that the A magnetization represents a TRM acquired during the cooling episode indicated by the ca. 2020 My mineral ages from the area. The Limpopo A pole is displaced about 20° along the APW path in an "older" direction from the Great Dyke extensions pole. Thus the palaeomagnetic results confirm the radiometric evidence that the more northerly part of the Limpopo belt in the vicinity of the Great Dyke extensions cooled appreciably later than the more southerly and westerly parts of the belt in South Africa and Botswana. Furthermore, the Great Dyke extensions and the Limpopo A magnetizations are of opposite polarity. The Great Dyke extensions pole shown in Fig. 7 is that corresponding to a south-seeking magnetization, all the samples investigated having southerly declinations and negative inclinations. The Limpopo A pole, however, is based almost exclusively on north-seeking directions – all the magnetizations, with the exception of the low coercivity components from site 16 (see below), having northerly positive directions. Thus during the time interval between the cooling of the two parts of the belt there was a change in geomag-

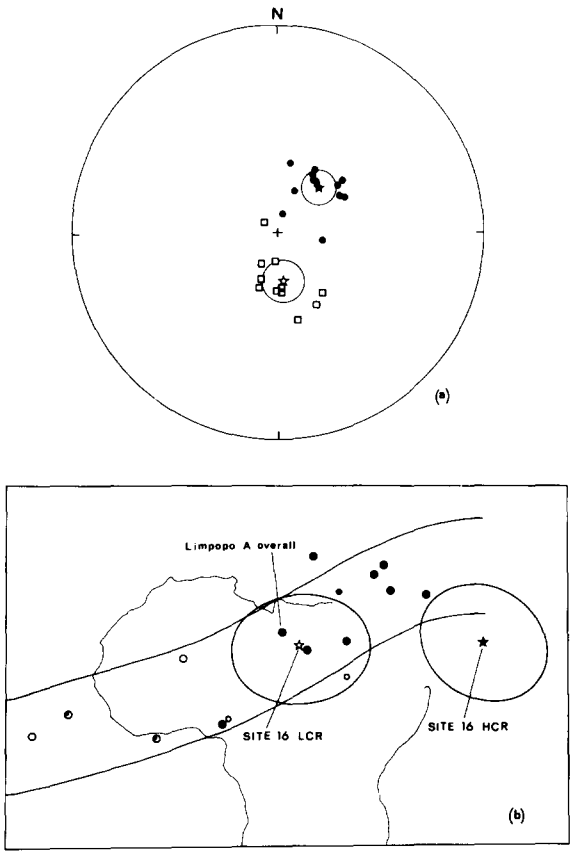


Fig. 10. Normal and reversed A components at Limpopo site 16. (a) Stereographic projection showing the HCR component – the observed magnetization after cleaning at 100 mT (circles, lower hemisphere), and the LCR component – the removed vectors at generally the 30/40 mT step in progressive demagnetization (squares, upper hemisphere). The two mean directions with their circles of 95% confidence are shown by stars. (b) Segment of the African APW path from Fig. 7 showing also the poles corresponding to the site 16 HCR and LCR magnetizations. Polarity convention as in Fig. 7.

netic field polarity: because the record of this event is preserved in what are almost certainly slowly acquired magnetizations, it is not possible to say at present whether this change was a reversal from one constant polarity to the other or merely a change in polarity bias. Thus there is the interesting possibility that further detailed sampling, especially in the Rhodesian part of the belt, might allow this polarity change to be mapped on the ground.

At site 16 in the northern part of the Kaapvaal craton both normal and reversed A components were found in every specimen (Fig. 10). A HCR (high coercivity of remanence) component with a fairly typical A direction was shown by the observed magnetizations after 100 mT demagnetization, after which step the magnetization had reached a stable end-point and had also achieved the best grouping. An examination of the vectors removed during the earlier stages of demagnetization, generally at the 30/40 mT step, showed that they were also well grouped, and that this LCR (low coercivity of remanence) component was imperfectly antiparallel to the HCR component. The palaeomagnetic poles corresponding to the HCR and LCR components fall approximately on older and younger parts of the APW path respectively, and so it seems likely that the imprecision of their reversal is due to APW having occurred between the times that the higher and lower coercivity fractions were magnetized. Thermal demagnetization of a single specimen from site 16 indicated that the removed vectors were predominantly in the same directions as the HCR and LCR components revealed by AF demagnetization, but that the LCR directions were associated with *higher* blocking temperatures than the HCR directions. Thus it may not be safe to assume that the apparently older HCR and younger LCR components were acquired simply as separate TRMs during slow cooling. Although there is obviously some uncertainty attached to pole positions from single sites, it would appear from their pole positions that the site 16 HCR magnetization was acquired considerably earlier than the average Limpopo A magnetization, and that the LCR magnetization was acquired close to the typical time. If this is so the reversed LCR magnetization might represent a rare record of a short reversed episode within a predominantly normal interval that is not seen at other Limpopo A sites because of the time-averaging effects of slow magnetization. Alternatively, the positions of the two poles may not accurately reflect their times of magnetization, perhaps due to localised tilting, and their reversal may be a reflection of the same polarity change that occurred between the typical Limpopo A and Great Dyke extension magnetizations.

At site 38 both A and B magnetizations were present in every specimen, and could be separated by detailed AF demagnetization and vector analysis. Figure 11 shows the site mean observed magnetizations

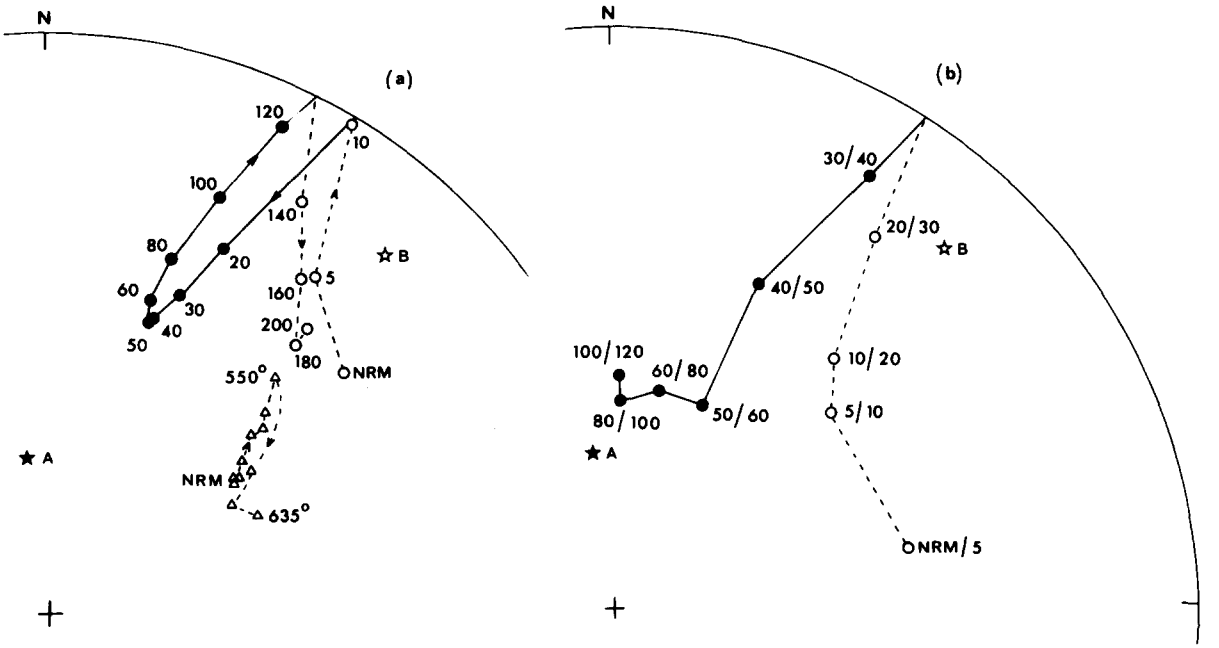


Fig. 11. Demagnetization of specimens from Limpopo site 38 illustrating coexisting A and B magnetizations in all specimens. (a) Circles, site mean observed magnetizations during AF demagnetization between NRM and 200 mT; triangles, observed magnetizations during thermal demagnetization of one specimen. (b) Site mean removed vectors up to 100/120 mT. Stars indicate overall mean A and B magnetizations from all sites. Stereographic projection, filled symbols lower hemisphere, open symbols upper hemisphere.

between the NRM stage and 200 mT, and also the site mean removed vectors between the NRM/5 mT and 100/120 mT steps. The observed NRM had a typical B direction in the upper hemisphere, but progressive AF demagnetization caused the magnetization to move systematically into the lower hemisphere towards the typical A direction and to reach an apparent end-point at 40 and 50 mT about 25° from the A direction. Further demagnetization caused the observed magnetization to move very regularly back into the upper hemisphere and reach an end-point at 180 and 200 mT close to the NRM direction. The direction of the vector removed in the NRM/5 mT step is of doubtful significance as it probably includes recently acquired components, but at successively higher steps the removed vectors moved gradually from the typical B direction towards the A direction, indicating that at first the B component alone was being removed, and then also an increasing proportion of A. The 60/80, 80/100 and 100/120 mT removed vectors appear to form an end-point reasonably close to the typical A direction, suggesting that over these steps the A component alone was being removed, although of course it is impossible to exclude completely the unlikely possibility that over these three steps the A and B components were removed in almost exactly similar proportions. The removed vectors above 120 mT were not well-grouped within each step, and the site means showed an erratic movement which for clarity is not illustrated, but which was generally scattered about the typical A direction. The gradual movement of the observed magnetization back to the B direction above 50 mT indicates the presence of a residual high-coercivity B component which is not being removed by demagnetization.

These results thus indicate that at site 38 the B component resides in grains with both high and low coercivities, while the A component resides in medium coercivity grains. Although there is an overlap of the coercivities of the two components between 20 and 60 mT, there is an interval between 60 and 120

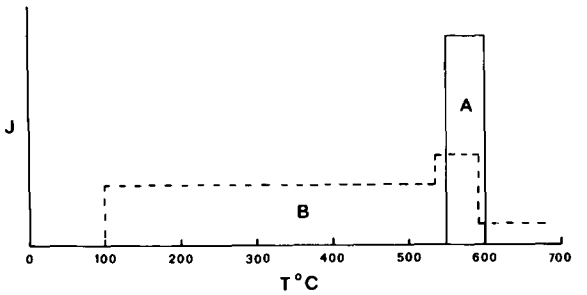


Fig. 12. Generalized blocking temperature spectra of the A and B magnetizations in the Limpopo belt gneisses.

mT where only the A component is removed, and it is this that allows the A component to be isolated. This situation is in contrast to that revealed by the thermal demagnetization of one specimen from site 38. Demagnetization up to 550° caused the observed magnetization to move only about 15° towards the A direction, and then between 550 and 615° the magnetization moved back to the NRM direction (Fig. 11). For all the demagnetization steps except two, the removed vectors (not illustrated) were close to the NRM direction, i.e. the B component alone was being removed. For the other two steps (550/585° and 585/615°) however, the removed vectors were displaced towards the A direction, but were still in the upper hemisphere more than 80° from the A direction, indicating that the B and A components were being removed together (in the ratio of approximately 2 : 1). Thus the A and B components at site 38 have completely overlapping blocking temperature spectra (similar to their spectra when they are present at different sites, see Fig. 12) and so thermal demagnetization alone cannot isolate the A component.

The B magnetization was found at eight sites, mainly from the central part of the belt in Botswana. At most of these sites the best within-site grouping was at the NRM stage, and preliminary thermal demagnetization indicates that the B magnetizations typically reside in grains with an unusually wide spectrum of blocking temperatures, from below 100 up to almost 700°C. The B magnetization is very similar to that reported for the Gaberones Granite of Botswana (Evans, 1967) both in its mean direction and its pattern of scatter. The granite has yielded ages which range between 1630 and 2980 My, but most of which are around 2300 My, which is generally thought to be the most probable age of intrusion (see review by Key, 1977). The remanence in the granite is thought to be primary, because the same direction is observed as an overprint in the contact zone of the Modipe Gabbro where it is intruded by the granite. Because of this it was originally thought, before thermal demagnetization results had been obtained, that the similarity between the Limpopo B and Gaberones magnetizations indicated that the Limpopo B magnetization was older than Limpopo A and had therefore survived the ca. 2000 My thermal event (Morgan, 1978). If the identification of the A magnetization as a TRM acquired during the ca. 2000 My event is correct, however, the characteristic blocking temperature spectra of the two magnetizations (Fig. 12) indicate that the original view is untenable. The B magnetization must be younger than A because its low blocking temperature elements have not been reset. Also B cannot be the result of a simple thermal overprint because it extends to higher blocking temperatures than A. The most likely explanation is that B is a low temperature CRM, chemical growth having produced grains with a very wide range of blocking temperatures. If the Gaberones Granite magnetization is primary then its similarity to the Limpopo B magnetization would appear to be fortuitous. In the absence of any post-2000 My geological event in the Limpopo belt with which to correlate it, the age of the B magnetization remains uncertain at present. It is noted, however, that the pole (reversed) corresponding to the B magnetization falls on the African APW path (Fig. 1b) at a point corresponding to an age of about 750 My.

The thermal demagnetization of the B magnetization in two specimens from widely separated sites is illustrated in Fig. 13. Between the NRM stage and 562°C, both showed very good directional stability, and their intensities were reduced to about 10% of their NRM values. Between 562 and 598°C however, i.e. on passing through the magnetite Curie point, the intensities suddenly dropped to about 1%, and there was a quite spectacular change in direction corresponding closely to a 180° reversal. Over the four steps between 598 and 691°C the intensities remained fairly level and the directions also stayed fairly constant in the reversed positions. For one of

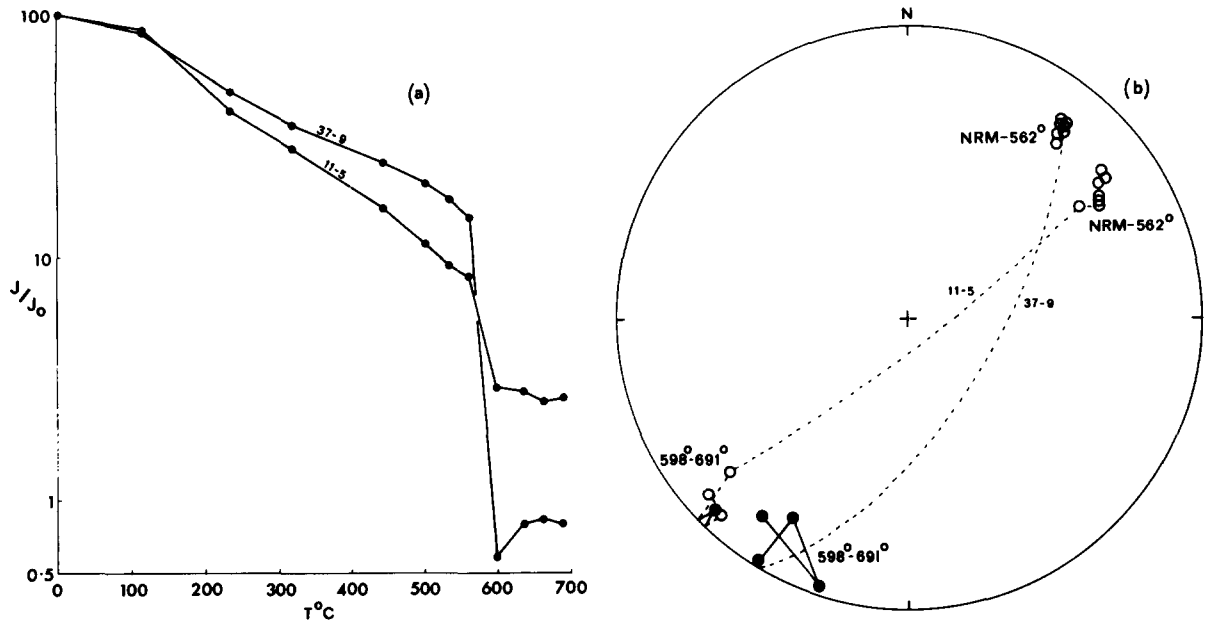


Fig. 13. Thermal demagnetization of two specimens with B magnetizations showing reversal during demagnetization. (a) Intensity decay curves; note logarithmic scale. (b) Equal area projection of directions. Filled circles lower hemisphere, open circles upper hemisphere.

these specimens (37-9) AF demagnetization of another specimen from the same core sample also produced the same reversal of direction, the change occurring between 60 and 80 mT. Whether these reversed remanences within single specimens are due to a reversal during slow chemical growth, or during slow cooling, or even represent a self-reversal within the magnetic phases, it is not known at present.

5.3. Discussion

The new palaeomagnetic results reported here, together with the previously published data for the interval ca. 2000–1850 My, result in a particularly well-defined APW path segment approximately 80° long and 20° wide (Fig. 7). It has been noted already that this APW path shows a clear polarity pattern, with the older eastern part of the path made up almost exclusively of magnetic N-poles, while the younger western part of the path is of mixed polarity but with a predominance of S-poles. The transition between the two polarity segments appears to occur between the Limpopo A and the Great Dyke extension poles. This may be the reason why, of all the major poles in the older group, the Limpopo A is the only one which contains a small element (the LCR component at site 16) of southerly magnetizations. It may also be significant that of the three major poles in the mixed polarity segment of the path, the Great Dyke extension pole is made up entirely of southerly magnetizations, whereas the Palabora 2 and Mashonaland Dolerites poles are mixed with a southerly bias. It is possible that the Great Dyke extension magnetizations, almost certainly acquired during regional slow cooling, is a time-averaged record of a geomagnetic field whose detailed behaviour is indicated by the spot-reading Palabora 2 and Mashonaland Dolerite magnetizations.

The change of polarity from almost exclusively N-poles to mixed but predominantly S-poles which is seen in the African APW path at about 1950 My is also seen in the Laurentian APW path at approximately the same time (Fig. 14). The Laurentian data suggests a long period, from about 2300 to 2000 My, characterized by almost exclusively one polarity, followed by an interval of perhaps 150–200 My of mixed polarity. Polarity–time patterns of this type could be a useful aid in the problem of the ambiguity

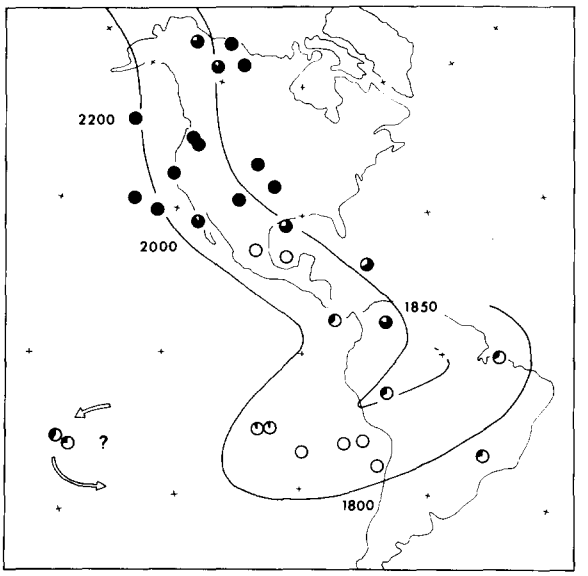


Fig. 14. APW path for Laurentia for interval ca. 2250 to 1750 My from McGlynn and Irving (1978). Filled circles, poles corresponding to N-seeking magnetizations; open circles, poles corresponding to S-seeking magnetizations; partly filled circles, both polarities with proportions indicated. Poles shown are those with N (sites) $\geqslant 3$ from listings given in McGlynn and Irving (1978), Irving and Naldrett (1977) and Roy and Lapointe (1976), for which polarity data could be obtained from references given.

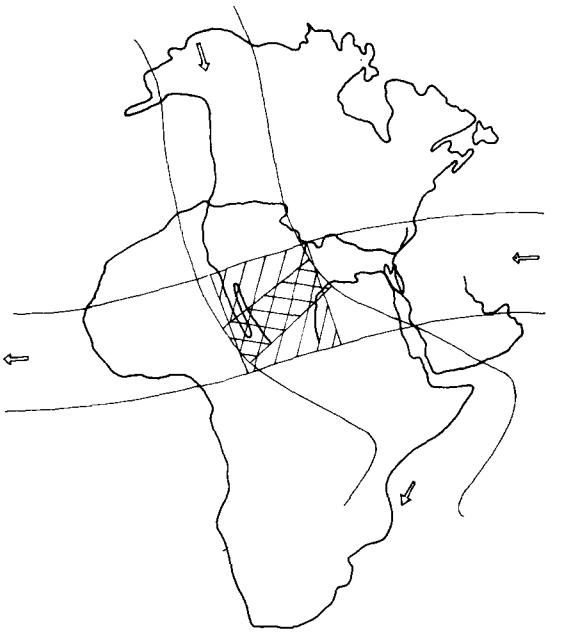


Fig. 15. Reconstruction showing possible configuration of Africa and Laurentia at about 1950 My. The relevant segments of the two APW paths are also shown, the shaded areas representing the regions in the two paths where single (north) polarity changes to mixed polarity. See text for further details.

of geographic polarity of Precambrian APW paths. For example, at present when comparing the APW paths of different continents, one cannot be certain that one is not comparing a N-geographic pole path with a S-geographic pole path. This ambiguity will not be completely resolved of course until continuity with well-established Phanerozoic paths is established. However, if these polarity–time patterns become sufficiently well-defined and distinctive in Precambrian APW paths then one could at least be sure of comparing like path segments with like. For example, if the APW paths from two different continents both show the same polarity pattern over the same time interval, then clearly we are comparing like segments when both have the same polarity poles in the same part of the pattern.

Both the African and Laurentian APW paths cross their parent continents at about 2000 My, and because the two continents could not have overlapped, it might be suggested that the two paths illustrated in Figs. 7 and 14 represent the traces of opposite geographic poles. However, the similarity in the polarity pattern of the two APW paths at about 2000–1850 My mentioned above strongly suggests that the paths drawn in Figs. 7 and 14 represent the trace of the same geographic pole, although of course we do not know which pole this is. Thus it seems very probable that Africa and Laurentia were very closely adjacent about 2000 My ago. A reconstruction showing one possible configuration of Africa and Laurentia is shown in Fig. 15. As the points on the two APW paths where they change from single to mixed polarity form a convenient common reference unaffected by minor dating errors, the reconstruction has been obtained by superposing the two paths at these points. Although many of the possible reconstructions using this reference point are invalidated by gross overlap problems, there are still many possible configurations where the overlap is smaller, and it is emphasized that the reconstruction shown in Fig. 15

is simply that in which the overlap appears to be minimal. The overlap of *present-day* continents does not of course invalidate a reconstruction if the overlapping areas did not exist as continental crust at the time under consideration. The amount of overlap that would be required to superpose the two APW paths over their whole lengths at ca. 2000–1850 My does, however, suggest that this option is unlikely. All viable options therefore have the two APW paths discordant and this has several interesting consequences as follows.

(1) Long term unity of the two continents is ruled out, and it follows that they were in relative motion.

(2) Any reconstruction (of which that in Fig. 15 is simply one possibility) is therefore only instantaneously valid, and the convergence and divergence of the two APW paths must be a reflection of collision and separation of the two continents.

(3) The "plate margins" inferred by this argument would be hard to find, but the igneous and metamorphic activity in South Africa and Rhodesia discussed in this paper might well be causally related effects in the "plate interior".

6. Conclusions

Conventional palaeomagnetic studies of unmetamorphosed Precambrian sedimentary and igneous rocks are simply an extension of Phanerozoic studies aimed at deciphering the major trends in palaeogeomagnetism and global tectonics but with uncertainties and imprecision which worsen with increasing age. Studies of metamorphic terrains, however, have other potential uses. As well as providing unusually precise details of the relevant segments of APW paths, they can also, in favourable circumstances, yield information about the cooling history and thermal conditions of these terrains. These are important parameters in the orogenic and thermal history of the crust and palaeomagnetism may have a uniquely important role to play in their analysis.

Acknowledgements

We wish to thank the management and Staff of the Palabora Mining Company, Rio Tinto South Africa Limited Pty., and the Phosphate Development Corporation for enabling our work at Palabora and providing facilities and assistance; the Director and officers of the Geological Survey of South Africa for their collaboration in sampling at Rust der Winter; Dr. R.M. Key of the Geological Survey of Botswana for advice on sampling localities in the Limpopo Belt in Botswana; Mr. R.E.P. Fripp of the Geology Department, University of the Witwatersrand, Johannesburg, for advice and help in field sampling in the Messina region, and Professor T.N. Clifford and staff of the Geology Department, University of the Witwatersrand, for their generous assistance with our work in general.

References

Baer, A.J., 1976. The Grenville Province in Helikian times: a possible model of evolution. Philos. Trans. R. Soc. London, Ser. A., 280: 499–515.

Balsley, J.R. and Buddington, A.F., 1957. Remanent magnetism of the Russell belt of gneisses, Northwest Adirondack Mountains, New York. Philos. Mag., Suppl. Adv. Phys., 6: 317–322.

Barton, J.M. and Ryan, B., 1977. A review of the geochronological framework of the Limpopo Mobile Belt. In: I.F. Ermanovics, R.M. Key and G. McEwen (Editors), The Proceedings of a Seminar Pertaining to the Limpopo Mobile Belt. Bull. Geol. Surv. Botswana, Lobatse, 12: 183–200.

Beckmann, G.E.J., 1976. A palaeomagnetic study of part of the Lewisian complex, north-west Scotland. J. Geol. Soc. London, 132: 45–59.

Beckmann, G.E.J. and Mitchell, J.G., 1976. Palaeomagnetic and geochronological work in central West Greenland. Earth Planet. Sci. Lett., 30: 269–280.

Beckmann, G.E.J., Olesen, N.Ø. and Sørensen, K., 1977. A palaeomagnetic experiment on crustal uplift in West Greenland. Earth Planet. Sci. Lett., 36: 269–279.

Bingham, D.K. and Evans, M.E., 1975. Precambrian geomagnetic field reversal. Nature (London), 253: 332–333.

Bingham, D.K. and Evans, M.E., 1976. Palaeomagnetism of the Great Slave Supergroup, Northwest Territories, Canada: the Stark Formation. Can. J. Earth Sci., 13: 563–578.

Briden, J.C., 1973. Applicability of plate tectonics to pre-Mesozoic time. Nature (London), 244: 400–405.

Briden, J.C., 1976. Application of palaeomagnetism to Proterozoic tectonics. Philos. Trans. R. Soc. London, Ser. A., 280: 405–416.

Briden, J.C., Morris, W.A. and Piper, J.D.A., 1973, Paleomagnetic studies in the British Caledonides – VI, regional and global implications. Geophys. J.R. Astron. Soc., 34: 107–134.

Buchan, K.L. and Dunlop, D.J., 1976. Palaeomagnetism of the Haliburton intrusions: superimposed magnetizations, metamorphism, and tectonics in the late Precambrian. J. Geophys. Res., 81: 2951–2967.

Burke, K., Dewey, J.F. and Kidd, W.S.F., 1976. Precambrian palaeomagnetic results compatible with contemporary operation of the Wilson Cycle. Tectonophysics, 33: 287–299.

Compston, W. and McElhinny, M.W., 1975. The Rb–Sr age of the Mashonaland Dolerites of Rhodesia and its significance for palaeomagnetic correlation in southern Africa. Precamb. Res., 2: 305–315.

Dodson, M.H. and McClelland-Brown, E., 1979. Magnetic blocking temperatures of single domain grains slow cooling. J. Geophys. Res., 85: 2625–2637.

Embleton, B.J.J., 1978. The palaeomagnetism of 2400 m.y. rocks from the Australian Pilbara craton and its relation to Archaean–Proterozoic tectonics. Precamb. Res., 6: 275–291.

Evans, M.E., 1967. A palaeomagnetic study of the Gaberones Granite of Botswana. Geophys. J. R. Astron. Soc., 12: 491–498.

Evans, M.E. and Bingham, D.K., 1976. Palaeomagnetism of the Great Slave Supergroup, Northwest Territories, Canada: the Tochatwi Formation. Can. J. Earth Sci., 13: 555–562.

Evans, M.E. and McElhinny, M.W., 1966. The palaeomagnetism of the Modipe Gabbro. J. Geophys. Res., 71: 6053–6063.

Gough, D.I. and Van Niekerk, C.B., 1959. A study of the palaeomagnetism of the Bushveld Gabbro. Philos. Mag., 4: 126–136.

Hanekom, H.J., Van Staden, C.M.v.H., Smit, P.J. and Pike, D.R., 1965. The geology of the Palabora igneous complex. Geol. Surv. S. Afr. Mem., 54: 1–179.

Hargraves, R.B., 1970. Palaeomagnetic evidence relevant to the origin of the Vredefort Ring. J. Geol., 78: 253–263.

Hargraves, R.B. and Burt, D.W., 1967. Palaeomagnetism of the Allard Lake anorthosite suite. Can. J. Earth Sci., 4: 357–369.

Harper, C.T., 1967. On the interpretation of potassium–argon ages from Precambrian shields and Phanerozoic orogens. Earth Planet. Sci. Lett., 3: 128–132.

Harrington, H.J., Burns, K.L. and Thompson, B.P., 1973. Gambier–Beaconsfield and Gambier–Sorell Fracture zones and the movement of plates in the Australia–Antarctica–New Zealand region. Nature (London) 245: 109–112.

Hickman, M.H. and Wakefield, J., 1975. Tectonic implications of new geochronologic data from the Limpopo belt at Pikwe, Botswana, southern Africa. Geol. Soc. Am. Bull., 86: 1468–1472.

Irving, E. and Hastie, J., 1975. Catalogue of Palaeomagnetic Directions and Poles. Second Issue. Precambrian Results 1957–1974. Geomagn. Ser. No. 3. Publ. Earth Phys. Branch, Energy, Mines and Resources, Ottawa, Ont.

Irving, E. and McGlynn, J.C., 1976a. Proterozoic magnetostratigraphy and the tectonic evolution of Laurentia. Philos. Trans. R. Soc. London, Ser. A, 280: 433–468.

Irving, E. and McGlynn, J.C., 1976b. Polyphase magnetization of the Big Spruce Complex, Northwest Territories. Can. J. Earth Sci., 13: 476–489.

Irving, E. and Naldrett, A.J., 1977. Palaeomagnetism in Abitibi Greenstone Belt, and Abitibi and Matachewan diabase dikes. J. Geol., 85: 157–176.

Irving, E. and Park, J.K., 1972. Hairpins and superintervals. Can. J. Earth Sci., 9: 1318–1324.

Irving, E. and Park, J.K., 1973. Palaeomagnetism of metamorphic rocks: errors owing to intrinsic anisotropy. Geophys. J.R. Astron. Soc., 34: 489–493.

Irving, E. and Pullaiah, G., 1976. Reversals of the geomagnetic field, magnetostratigraphy, and relative magnitude of palaeosecular variation in the Phanerozoic. Earth Sci. Rev., 12: 35–64.

Irving, E. and Runcorn, S.K., 1957. Analysis of the palaeomagnetism of the Torridonian Sandstone Series of Northwest Scotland. Philos. Trans. R. Soc. London, Ser. A, 250: 83–99.

Irving, E., Emslie, R.F. and Ueno, H., 1974. Upper Proterozoic palaeomagnetic poles from Laurentia and the history of the Grenville Province. J. Geophys. Res., 79: 5491–5502.

Irving, E., Tanczyk, E. and Hastie, J., 1976a. Catalogue of Palaeomagnetic Directions and Poles. Third issue. Palaeozoic Results 1949–1975. Geomagn. Ser. No. 5. Publ. Earth Phys. Branch, Energy, Mines and Resources, Ottawa, Ont.,

Irving, E., Tanczyk, E. and Hastie, J., 1976b. Catalogue of Palaeomagnetic Directions and Poles. Fourth issue. Mesozoic Results 1954–1975. Geomagn. Ser. No. 6. Publ. Earth Phys. Branch, Energy, Mines and Resources, Ottawa, Ont.

Irving, E., Tanczyk, E. and Hastie, J., 1976c. Catalogue of Palaeomagnetic Directions and Poles. Fifth issue. Cenozoic Results 1927–1975. Geomagn. Ser. No. 10. Publ. Earth Phys. Branch, Energy, Mines and Resources, Ottawa, Ont.

Jones, D.L. and McElhinny, M.W., 1967. Stratigraphic interpretation of palaeomagnetic measurements on the Waterberg redbeds of South Africa. J. Geophys. Res., 72: 4171–4179.

Jones, D.L., Robertson, I.D.M. and McFadden, P.L., 1975. A palaeomagnetic study of Precambrian dyke swarms associated with the Great Dyke of Rhodesia. Trans. Geol. Soc. S. Afr., 78: 57–65.

Key, R.M., 1977. The geochronology of Botswana. Trans. Geol. Soc. S. Afr., 80: 31–42.

McElhinny, M.W., 1968. Palaeomagnetic directions and pole positions – IX. Geophysics J. R. Astron. Soc., 16: 207–224.

McElhinny, M.W., 1973. Palaeomagnetism and Plate Tectonics. Cambridge University Press, London, 357 pp.

McElhinny, M.W. and McWilliams, M.O., 1977. Precambrian geodynamics – a palaeomagnetic view. Tectonophysics, 40: 137–159.

McElhinny, M.W. and Opdyke, N.D., 1964. The palaeomagnetism of the Precambrian dolerites of eastern Southern Rhodesia, an example of geologic correlation by rock magnetism. J. Geophys. Res., 69: 2465–2475.

McElhinny, M.W., Briden, J.C., Jones, D.L. and Brock, A., 1968. Geological and geophysical implications of palaeomagnetic results from Africa. Rev. Geophys., 6: 201–238.

McGlynn, M.W. and Irving, E., 1978. Multicomponent magnetization of the Pearson Formation (Great Slave Supergroup, N.W.T.) and the Coronation loop. Can. J. Earth Sci., 15: 642–654.

McWilliams, M.O., 1977. Late Precambrian palaeomagnetism of Austrlia and Africa. Ph.D. thesis, Australian National University, Canberra, A.C.T.

McWilliams, M.O. and Dunlop, D.J., 1978. Grenville palaeomagnetism and tectonics. Can. J. Earth Sci., 15: 687–695.

McWilliams, M.O., McElhinny, M.W. and Kroner, A., 1978. Late Precambrian palaeomagnetism of Australia and Africa; for how long did Gondwanaland exist? Eos Trans. Am. Geophys. Union, 59: 265 (abstract).

Morgan, G.E., 1976. Palaeomagnetism of a slowly cooled plutonic terrain in western Greenland. Nature (London), 259: 382–385.

Morgan, G.E., 1978. Preliminary palaeomagnetic results from the Limpopo mobile belt. Geophys. J. R. Astron. Soc., 53: 152–153 (abstract).

Oosthuyzen, E.J., 1964. 'n geochronologiese studie van sekere stollingsgesteentes net over en jonger as die sisteern Waterberg. M.Sc. thesis, University of Pretoria,

Park, J.K., 1973. Palaeomagnetism of metamorphic rocks of Daly Bay and Melville Peninsula, Churchill Structural Province. Can. J. Earth Sci., 10: 1079–1088.

Park, J.K., 1975. Palaeomagnetism of the Flin Flon – Snow Lake greenstone belt, Manitoba and Saskatchewan. Can. J. Earth Sci., 12: 1272–1291.

Park, J.K., 1977. A reconnaissance palaeomagnetic survey of the central Labrador Trough, Quebec. Can. J. Earth Sci., 14: 159–174.

Piper, J.D.A., 1976a. Palaeomagnetic evidence for a Proterozoic super-continent. Philos. Trans. R. Soc. London, Ser. A, 280: 469–490.

Piper, J.D.A., 1976b. Definition of pre-2000 m.y. apparent polar movements. Earth Planet. Sci. Lett., 28: 470–478.

Piper, J.D.A., Briden, J.C. and Lomax, K., 1973. Precambrian Africa and South America as a single continent. Nature (London), 245: 244–248.

Poorter, R.P.E., 1972. Palaeomagnetism of the Rogaland Precambrian (south-western Norway). Phys. Earth Planet. Inter., 5: 167–176.

Poorter, R.P.E., 1975. Palaeomagnetism of Precambrian rocks from southeast Norway and south Sweden. Phys. Earth Planet. Inter., 10: 74–87.

Pullaiah, G., Irving, E., Buchan, K.L. and Dunlop, D.J., 1975. Magnetization changes caused by burial and uplift. Earth Planet. Sci. Lett., 28: 133–143.

Roy, J.L. and Lapointe, P.L., 1976. The palaeomagnetism of Huronian red beds and Nipissing diabase, post Huronian events and apparent polar path for the interval 2300 to 1500 Ma for Laurentia. Can. J. Earth Sci., 13: 749–773.

Sabaté, P. and Lomax, K., 1975. Données stratigraphiques et paléomagnétiques de la région Yetti-Eglab (Sahara occidental algérien). Bull. Bur. Rech. Geol. Minieres Fr., Sect. 2: 293–311.

Van Breemen, O. and Dodson, M.H., 1972. Metamorphic chronology of the Limpopo Belt, southern Africa. Bull. Geol. Soc. Am., 83: 2005–2018.